16/06/2026

Code-P

Aakash
Medical|IIT-JEE|Foundations

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MM : 720 **Project Phoenix for Re-NEET-2026_CST-10P** **Time : 180 Min.**

Topics Covered:
Complete Syllabus of Class XI & XII

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are three parts in the question paper consisting of Physics, Chemistry and Biology which have 45 questions each in Physics & Chemistry and 90 questions in Biology.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be

(1) 524 Hz

(2) 536 Hz

(3) 537 Hz

(4) 523 Hz

2. An equiconvex lens has refractive index μ and radius of curvature of each surface is R. It is cut into two symmetrical halves by a plane perpendicular to the principal axis. The power of one part is

(1) $\frac{\mu-1}{2R}$

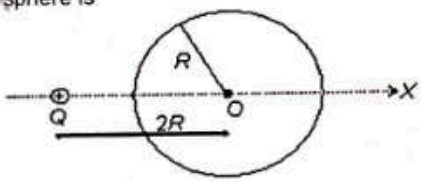
(2) $\frac{\mu-1}{R}$

(3) $\frac{2(\mu-1)}{R}$

(4) $\frac{2\mu-1}{R}$

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3. An uncharged metallic solid sphere of radius R is placed at a distance $2R$ from point charge Q as shown in figure. The electric field intensity due to induced charge at centre of sphere is



(1) Zero

(2) $\frac{Q}{2\pi\varepsilon_0 R^2}$ towards left

(3) $\frac{Q}{16\pi\varepsilon_0 R^2}$ towards right

(4) $\frac{Q}{16\pi\varepsilon_0 R^2}$ towards left

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4. If horizontal range is 8 times the maximum height attained by the projectile, then the angle of projection with the horizontal will be

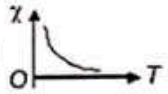
(1) $\tan^{-1}(2)$

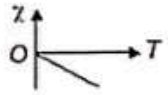
(2) $\cot^{-1}(2)$

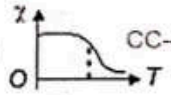
(3) $\sin^{-1}\left(\frac{1}{2}\right)$

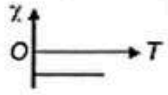
(4) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

5. The variation of magnetic susceptibility (χ) with absolute temperature (T) for a diamagnetic material is

(1) 

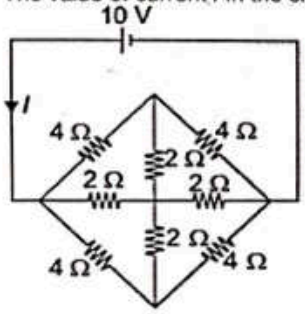
(2) 

(3) 

(4) 

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6. The value of current I in the circuit diagram shown will be



(1) 2 A

(2) 4 A

(3) 5 A

(4) 6 A

7. A circular loop of area 2.4 cm^2, carrying current 4 A is placed in a magnetic field of 0.1 T perpendicular to the plane of loop. The torque acting on the loop will be

(1) Zero

(2) 9.6×10^{-4} N m

(3) 4.8×10^{-2} N m

(4) 9.6×10^{-6} N m

8. The position co-ordinates of an object are given by $x = 2t^2$, $y = t^2 - 4t$. The average velocity of particle in first 1 s is

(1) 10 unit

(2) 12 unit

(3) $\sqrt{13}$ unit

(4) 8 unit

9. A lead sphere of mass m falls in a viscous liquid with terminal velocity v_0. Another lead sphere of mass M falls through with terminal velocity $4v_0$. The ratio $\frac{M}{m}$ is

(1) 2

(2) 4

(3) 8

(4) 16

10. A string of length 'l' is fixed at both ends. It is vibrating in its 3rd overtone with maximum amplitude 'a'. The amplitude at a distance $\frac{l}{3}$ from one end is

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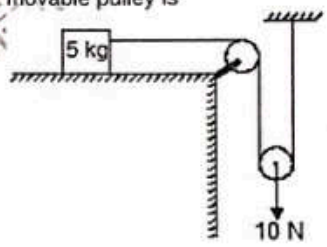
(1) a

(2) Zero

(3) $\frac{\sqrt{3}a}{2}$

(4) $\frac{a}{2}$

11. In the figure pulley and strings are massless and frictionless. The horizontal surface is smooth. The acceleration of the movable pulley is



(1) 2 m/s^2

(2) 1 m/s^2

(3) 0.5 m/s^2

(4) 4 m/s^2

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12. Read the statements carefully and choose the correct option
Statement A: A moving coil galvanometer can be converted into an ammeter by introducing a resistance of the small value in series.
Statement B: A moving coil galvanometer can be converted into a voltmeter by introducing a resistance of a large value in series.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Neither statement A nor B is correct

13. The power in ac circuit is given by $P = E_{rms}\, I_{rms} \cos\phi$. The value of $\cos\phi$ in series LCR circuit at resonance is

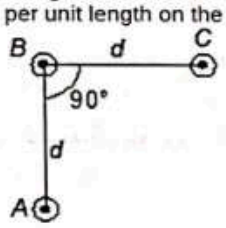
(1) Zero

(2) 1

(3) $\frac{1}{2}$

(4) $\frac{1}{\sqrt{2}}$

14. An arrangement of three parallel straight wires placed perpendicular to plane of paper carrying same current 'I' along the same direction is shown in Fig. Magnitude of force per unit length on the middle wire 'B' is given by



(1) $\frac{\mu_0 I^2}{2\pi d}$

(2) $\frac{2\mu_0 I^2}{\pi d}$

(3) $\frac{\sqrt{2}\mu_0 I^2}{\pi d}$

(4) $\frac{\mu_0 I^2}{\sqrt{2}\pi d}$

15. Two bodies of masses m and M are placed at distance d apart. The gravitational potential (V) at the position where the gravitational field due to them is zero is

(1) $V = -\frac{G}{d}$

(2) $V = -\frac{GM}{d}$

(3) $V = -\frac{G}{d}(\sqrt{m} + \sqrt{M})^2$

(4) $V = -\frac{G}{d}(m + M)$

16. Two particles executing SHM along x-axis have same frequency and same amplitude, meet at $x = \frac{A}{2}$, while moving in opposite direction. Phase difference between particles is

(1) $\frac{\pi}{3}$

(2) $\frac{2\pi}{3}$

(3) $\frac{5\pi}{6}$

(4) $\frac{\pi}{6}$

17. Binding energy for ${}_6C^{12}$ is 92.16 MeV, for ${}_3Li^7$ is 39.3 MeV and for ${}_2He^4$ is 27.37 MeV. Then stability order will be

(1) $C^{12} > Li^7 > He^4$

(2) $Li^7 > He^4 > C^{12}$

(3) $C^{12} > He^4 > Li^7$

(4) $He^4 > Li^7 > C^{12}$

18. The dimensional formula of physical quantity X is $[ML^2T^{-2}]$. If maximum percentage error in M, L and T are 2%, 1% and 1% respectively, then the maximum percentage error in x is

(1) 4%

(2) 3%

(3) 6%

(4) 5%

19. If the electric field component of an electromagnetic wave moving in positive x-direction is given by $\vec{E} = 6\cos\left[1.2x - 3.6\times10^8 t\right]\hat{j}\ \frac{N}{C}$, then the average energy density of the electromagnetic wave will be

(1) 2.8×10^{-10} J/m^3

(2) 1.6×10^{-10} J/m^3

(3) 9×10^{10} J/m^3

(4) 3.8×10^{-8} J/m^3

20. A light has amplitude A after passing through the polariser. Angle between analyser and polariser being 60°. Light transmitted by analyser has amplitude

(1) $A\sqrt{2}$

(2) $\frac{A}{\sqrt{2}}$

(3) $\frac{\sqrt{3}A}{2}$

(4) $\frac{A}{2}$

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21. Match **List-I** with **List-II** (where symbols have their usual meaning).

List-I (Quantity)	List-II (Corresponding formula)
A. Torque τ	I. $M\vec{a}_{cm}$
B. Angular momentum	II. Iα
C. $\sum \vec{F}_{ext}$	III. Mvr⊥
D. Instantaneous rotational power	IV. $\vec{\tau}\cdot\vec{\omega}$

Chose the correct answer from the options given below:

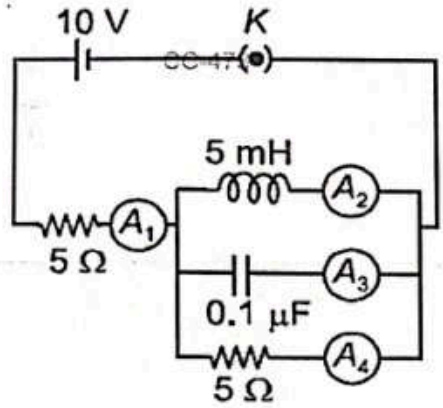
(1) A-III, B-II, C-I, D-IV

(2) A-III, B-II, C-IV, D-I

(3) A-II, B-III, C-I, D-IV

(4) A-II, B-III, C-IV, D-I

22. In the circuit shown after closing the key, in steady state, the readings of ammeters A_1, A_2, A_3 and A_4 are respectively



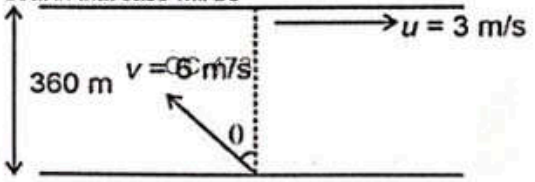
(1) 2 A, 0, 0, 2 A

(2) 1 A, 0, 0, 1 A

(3) 2 A, 2 A, 0, 0

(4) 1 A, 1 A, 0, 1 A

23. A boat can be moved at 6 m/s in still water. It is used to cross a 360 m wide river. The river current has uniform velocity of 3 m/s. The value of θ in which the boat should be steered to cross the river in minimum time and the drift of boat in that case will be



(1) θ = 0°, drift = 180 m

(2) θ = 30°, drift = 200 m

(3) θ = 45°, drift = 120 m

(4) θ = 90°, drift = 100 m

24. In a compound microscope, focal length of objective and eyepiece are 2 cm and 5 cm respectively. An object is placed at a distance 2.2 cm from objective. If final image is formed at least distance of distinct vision (25 cm), then magnifying power of microscope and approximate distance between objective and eyepiece will be respectively

(1) 60, 18 cm

(2) 60, 26 cm

(3) 50, 20 cm

(4) 50, 26 cm

25. Separation between electric dipole of dipole moment p and point charge q kept along axis of dipole is r, then magnitude of force on dipole due to charge is

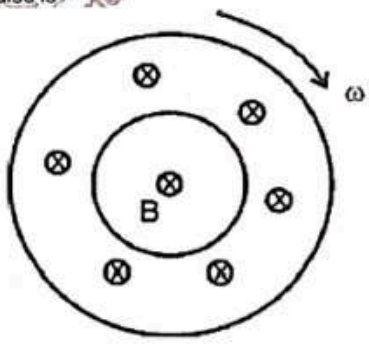
(1) 0

(2) $\frac{kpq}{r^2}$

(3) $\frac{kpq}{r^3}$

(4) $\frac{2kpq}{r^3}$

26. An annular disc of copper with inner and outer radius R and $2R$ is rotating with a uniform angular speed ω in a region where a uniform magnetic field B exist as shown. The net emf induced between the inner edge and outer edge of the disc is



(1) $\frac{B\omega R^2}{2}$

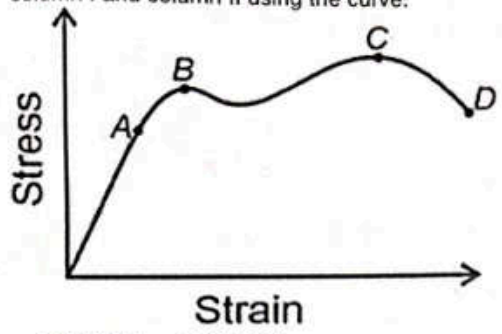
(2) $\frac{3B\omega R^2}{2}$

(3) $2B\omega R^2$

(4) Zero

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27. Stress-strain curve for a metal is shown below. Match the column I and column II using the curve.



Column-I	Column-II
a. Point A	(i) Yield point
b. Point B	(ii) Fracture point
c. Point C	(iii) Proportional limit
d. Point D	(iv) Ultimate strength

Choose the correct option.

(1) a(i), b(iii), c(ii), d(iv)

(2) a(i), b(iv), c(iii), d(ii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iii), b(i), c(iv), d(ii)

28. A : An ideal gas expands isothermally, during this process it absorbs 20 J heat. According to first law of thermodynamics work done by the gas is –20 J.
R : There will be no change in the internal energy of gas when work done on the gas is negative.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

29. A street performer is showing a feat in a busy lane of Punjabi Bagh, Delhi. He is carrying a monkey on his head and is cycling on a circular track of radius 6 m and completes 180 revolutions in 15 minutes. If total mass of man, monkey and cycle is 100 kg, then value of friction force on the system is nearly (Take, $\pi^2 \approx 10$)

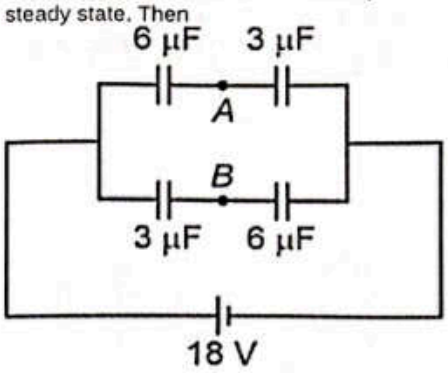
(1) 480 N

(2) 1120 N

(3) 280 N

(4) 960 N

30. In the circuit shown below the battery is ideal and circuit is in steady state. Then



(1) Potential difference $V_A - V_B$ is 6 V

(2) Potential difference $V_A - V_B$ is – 6 V

(3) Total charge supplied by battery is 36 μC

(4) Total charge supplied by battery is 84 μC

31. The displacement of a particle moving in SHM at any instant is given by $y = a\sin(\omega t)$. The velocity after time $t = \frac{T}{4}$ is (T is the time period)

(1) $a\omega^2$

(2) $-a\omega$

(3) Zero

(4) $a\omega$

32. The minimum orbital angular momentum of electron in a hydrogen atom is

(1) $\frac{h}{\pi}$

(2) $\frac{h}{2}$

(3) $\frac{h}{2\pi}$

(4) h

33. In a p-n junction diode, holes diffuse from p-region to n-region because

(1) There is large concentration of holes in p-region as compared to n-region

(2) There is large concentration of electrons in p-region as compared to the n-region

(3) The concentration of holes and electron in both region (p-region & n-region) is same.

(4) The mobility of holes is more than the mobility of electrons

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34. In YDSE, a thin glass slab ($\mu = 1.5$) is introduced in the path of one of the two interfering beams. The central fringe moves to a position occupied by the fifth bright fringe earlier. Thickness of the glass slab is ($\lambda = 5000$ Å)

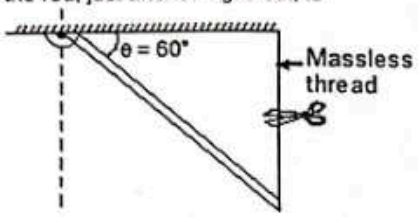
(1) 4000 Å

(2) 6000 Å

(3) 2000 Å

(4) 50000 Å

35. A rod of length 2 m hinged with ceiling can swing freely in vertical plane. Initially, it is held with the help of thread at 60° with horizontal as shown in figure. Angular acceleration of the rod, just after string is cut, is



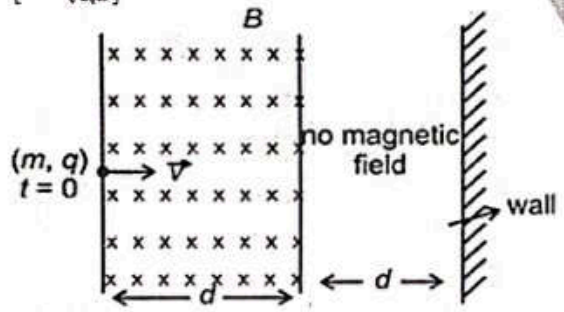
(1) 5.0 rad s^{-2} CC-473

(2) 2.0 rad s^{-2}

(3) 3.75 rad s^{-2}

(4) 4.25 rad s^{-2}

36. A particle of mass m and charge q enters a region of magnetic field with speed v as shown in the figure. If collision of particle with wall is elastic then time after which velocity of particle becomes antiparallel to its initial velocity is $\left[d = \frac{mv}{\sqrt{2}qB}\right]$



(1) $\frac{m}{2qB}(\pi+4)$

(2) $\frac{m}{4qB}(\pi+2)$ CC-473

(3) $\frac{m}{qB}(\pi+2)$

(4) $\frac{m}{4qB}(\pi-2)$

37. Certain quantity of water cools from 70°C to 60°C in the first 5 minutes and to 54°C in the next 5 minutes. The temperature of the surroundings is

(1) 45°C

(2) 20°C

(3) 42°C

(4) 10°C

38. **Statement A:** The value of $\frac{C_V}{C_P}$ for a monoatomic gas will be less as compared to that of a diatomic gas. (Symbols have their usual meanings)
Statement B: The molecules of a diatomic gas have more degree of freedom than those of monoatomic gas.

(1) Both statement (A) and statement (B) are correct

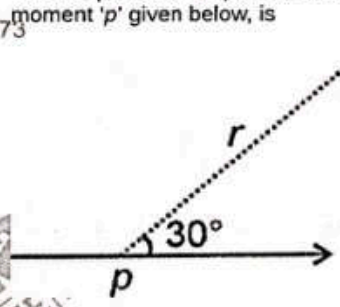
(2) Both statement (A) and statement (B) are incorrect

(3) Statement (A) is correct and statement (B) is incorrect

(4) Statement (A) is incorrect and statement (B) is correct

39. Electric potential at point A due to a dipole having dipole moment 'p' given below, is

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(1) $\frac{kp}{2r^2}$

(2) $\frac{\sqrt{3}kp}{2r^2}$

(3) $\frac{kp}{r^2}$

(4) $\frac{1}{\sqrt{2}}\frac{kp}{r^2}$

40. An electromagnetic radiation of 6.2 eV falls on a metal surface of work function ϕ = 4.2 eV. The K.E. (in joule) of the fastest electron emitted is nearly

(1) 3.2×10^{-21}

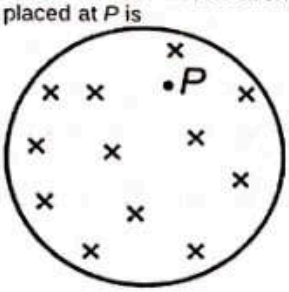
(2) 3.2×10^{-20}

(3) 3.2×10^{-19}

(4) 3.2×10^{-16}

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41. Figure shows a uniform magnetic field B confined to a cylindrical volume and is increasing at a constant rate. The instantaneous acceleration experienced by an electron placed at P is



(1) Zero

(2) Towards right

(3) Towards left

(4) Upwards

42. Read the following statements.
(a) The angle of contact of mercury with glass is obtuse.
(b) Surface tension of a liquid is independent of the area of surface.
(c) On adding some detergent to water, the surface tension of water increases.
(d) A drop of liquid under no external force is always spherical in shape.
The **incorrect** statement(s) is/are

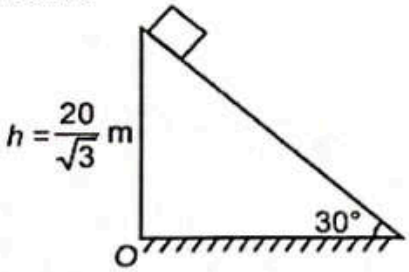
(1) (a) and (c) only

(2) (a) and (b) only

(3) (c) and (d) only

(4) (c) only

43. If a block of mass 5 kg starts sliding on a fixed smooth inclined plane at $t = 0$ from height $\frac{20}{\sqrt{3}}$ m as shown in figure. Its angular momentum about a point O after 1 s from release would be



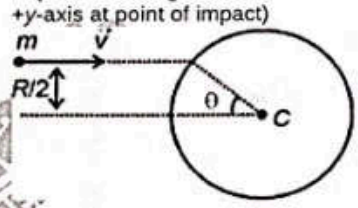
(1) 150 kg m^2s^{-1}

(2) 200 kg m^2s^{-1}

(3) 250 kg m^2s^{-1}

(4) 300 kg m^2s^{-1}

44. A particle of mass m strikes elastically with a disc of same mass with speed v as shown in figure. All the surface of contacts are smooth. The velocity of disc just after collision is (consider tangential line as $+x$-axis and radial toward C as $+y$-axis at point of impact)



(1) $\frac{\sqrt{3}}{2}v\hat{j}$

(2) $\frac{\sqrt{3}}{2}v\hat{i}$

(3) $\frac{v}{2}\hat{j}$

(4) $\frac{v}{2}\hat{i}$

45. The energy (E) and momentum (p) of photon are respectively; (symbols have their usual meaning)

(1) $E = hc, p = \frac{\nu}{c}$

(2) $E = h\nu, p = \frac{h\nu}{c}$

(3) $E = \frac{c}{\nu}, p = \frac{h}{c}$

(4) $E = \frac{h\nu}{\nu}, p = h\nu$

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CHEMISTRY

46. If 2 g of caustic soda (NaOH) is treated with 50 mL of 0.50 M HCl solution, then mass of the reactant left will be

(1) 0.8 g NaOH

(2) 1.0 g NaOH

(3) 0.9 g HCl

(4) 0.45 g HCl

47. Select the correct increasing order of energy for the following transition of a photon of hydrogen atom.

(1) (6 → 5) < (5 → 3) < (5 → 4) < (6 → 1)

(2) (6 → 5) < (5 → 4) < (5 → 3) < (6 → 1)

(3) (5 → 4) < (6 → 5) < (5 → 3) < (6 → 1)

(4) (6 → 1) < (5 → 3) < (5 → 4) < (6 → 5)

48. For Na atom, how many orbital(s) would have n + l = 2?

(1) 2

(2) 3

(3) 1

(4) 4

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49. Consider the following statements :
(I) First line of Lyman series have larger wavelength than second line of Lyman series
(II) The number of degenerate orbitals in 3rd shell in H-atom is 3
(III) 4*s* orbital is filled prior to 3*d* orbital
The incorrect statement(s) is/are

(1) Only II

(2) I, II

(3) Only III

(4) II, III

50. Number of cyclic ethers excluding stereoisomers for molecular formula C_4H_8O is

(1) 3

(2) 4

(3) 5

(4) 6

51. In the following reaction

$$C_2H_4 \xrightarrow[CCl_4]{Br_2} (A) \xrightarrow{\text{Alc. KOH}} (B) \xrightarrow{NaNH_2} \underset{\text{Major}}{(C)}$$

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Incorrect statement about product (C) is

(1) It is less acidic than ethane

(2) It reacts with water on warming with mercuric sulphate and dilute sulphuric acid at 333 K to form ethanal

(3) It reacts with two molecules of hydrogen bromides to form gem dibromides

(4) It reacts with $NaNH_2$ to liberate NH_3 gas

52. Read the statements marked as assertion (A) and reason (R).
Assertion (A): For isothermal reversible change work done by the system is equal to change in internal energy.
Reason (R): Change in internal energy depends on temperature of system.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

53. Consider the given redox reaction in acidic medium

$$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$$

The number of moles of $KMnO_4$ required to oxidise 0.5 mol of oxalate ions is

(1) 0.1 mol

(2) 0.2 mol

(3) 0.5 mol

(4) 1 mol

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54. If solubility of $Al_2(SO_4)_3$ is 's' mol L^{-1}, then the solubility product (K_{sp}) is

(1) $72s^5$

(2) $6s^2$

(3) $54s^4$

(4) $108s^5$

55. If equilibrium constant of a reaction at 27°C is 10 then the standard Gibbs free energy change for a reaction at equilibrium will be

(1) –3.54 kcal mol^{-1}

(2) –2.12 kcal mol^{-1}

(3) –1.38 kcal mol^{-1}

(4) Zero

56. In Dumas method for estimation of nitrogen, 0.1 g of an organic compound gave 30 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound is
(Aqueous tension at 300 K = 15 mm Hg)

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(1) 19.7%

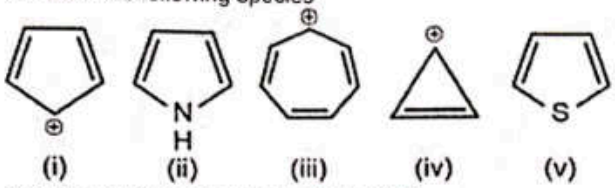
(2) 49.1%

(3) 28.2%

(4) 31.4%

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57. Consider the following species



(i) (ii) (iii) (iv) (v)

The total number of aromatic species are

(1) 2

(2) 4

(3) 5

(4) 3

58. Match the species given in column I with the shape given in column II and find the correct option

	Column I		Column II
a.	XeF_4	(i)	Linear
b.	I_3^-	(ii)	Pyramidal
c.	PCl_3	(iii)	Square planar
d.	NO_2^+	(iv)	Bent

(1) a(iv), b(i), c(ii), d(iii)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(ii), b(iv), c(i), d(iii)

59. In which of the following compounds, (C – C) bond rotation does not take place?

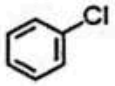
(1) C_4H_{10}

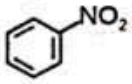
(2) C_2H_6

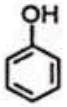
(3) C_3H_8

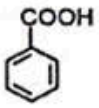
(4) C_2H_4

60. Which of the following organic compounds is most reactive towards electrophilic substitution reaction?

(1) 

(2) 

(3) 

(4) 

61. A compound with molecular formula of C_6H_{14} has two secondary and one tertiary carbons. Its IUPAC name is

(1) n-Hexane

(2) 3-Methylpentane

(3) 2,3-Dimethylbutane

(4) 2,2-Dimethylbutane

62. Given below are two statements

Statement I: In Br_3O_8, central bromine atom has +6 oxidation state.

Statement II: $CH_4(g) + 2O_2(g) \xrightarrow{\Delta} CO_2(g) + 2H_2O$ (l) is an example of decomposition redox reaction.

In the light of above statements, choose the correct answer.

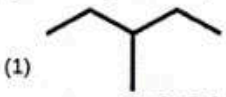
(1) Both statement I and statement II are correct

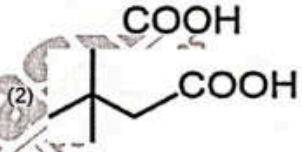
(2) Both statement I and statement II are incorrect

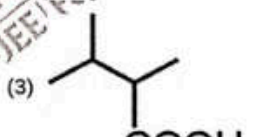
(3) Statement I is correct but statement II is incorrect

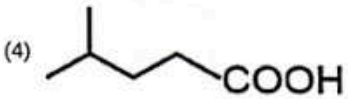
(4) Statement I is incorrect but statement II is correct

63. Sodium salt of which of the carboxylic acids on heating with soda lime gives neopentane as major products?

(1) 

(2) 

(3) 

(4) 

64. The entropy change when 1 mole of ideal gas at 27°C is expanded reversibly from 2 litre to 20 litre is

(1) 2 cal K^{-1} mol^{-1}

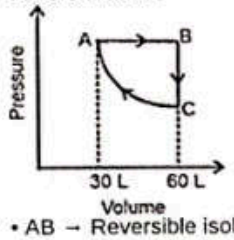
(2) 4.6 cal K^{-1} mol^{-1}

(3) 4 J K^{-1} mol^{-1}

(4) 4.6 kJ K^{-1} mol^{-1}

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65. 3 moles of a monoatomic ideal gas undergo following transformations



- AB → Reversible isobaric expansion
- BC → Reversible isochoric process
- CA → Reversible isothermal compression at 27°C

The work done for the process C → A and the value of change in enthalpy for overall process respectively are

(1) 51.1 kJ and zero

(2) 51.1 kJ and 1.8 cal

(3) 5.17 kcal and 1.8 cal

(4) 1.24 kcal and zero

66. Which species has 90° bond angle(s)?

(1) PCl_3F_2

(2) $SiCl_4$

(3) IF_5

(4) CO_3^{2-}

67. Arrange the given aqueous solutions in correct order of freezing point.
(a) 0.1 m urea
(b) 0.1 m NaCl
(c) 0.1 m $CaCl_2$
The correct option is

(1) (a) < (b) < (c)

(2) (b) < (c) < (a)

(3) (c) < (b) < (a)

(4) (c) < (a) < (b)

68. Which of the following is/are the application(s) of Kohlrausch law?
(i) Λ_m° for any electrolyte can be calculated by using λ_m° of individual ions
(ii) Degree of dissociation of weak electrolyte can be determined
(iii) Dissociation constant of weak electrolytes can be determined.
Choose the correct statements.

(1) (i) and (ii) only

(2) (ii) and (iii) only

(3) (i) and (iii) only

(4) (i), (ii) and (iii)

69. If the hydrogen electrode is dipped in an aqueous solution of pH = 2 at 25°C and 1 bar pressure. The reduction potential of the electrode would be

(1) -0.0591 V

(2) +0.0591 V

(3) +0.118 V

(4) -0.118 V

70. For an elementary reaction, A + 2B → C
If volume of the vessel is reduced to $\frac{1}{3}$ of its original volume, the rate of the reaction will

(1) Increase 27 times

(2) Remain unaltered

(3) Increase 9 times

(4) Decrease 3 times

71. If a first order reaction completed to 75% in 6 hours the velocity constant (in hr^{-1}) for the reaction is

(1) 0.693/6

(2) 6 × 0.693

(3) 0.693/3

(4) 3 × 0.613

72. Match list-I with list-II

	List-I		List-II
a.	$[Cr(NH_3)_4Br_2]Br$	(i)	Diamagnetic and exhibits cis-trans isomerism
b.	$[Ti(H_2O)_5Br](NO_3)_2$	(ii)	Diamagnetic and exhibits ionization isomerism
c.	$[Pt(en)(NH_3)Br]NO_3$	(iii)	Paramagnetic and exhibits cis-trans isomerism
d.	$[Co(NH_3)_4(NO_3)_2]NO_3$	(iv)	Paramagnetic and exhibits ionization isomerism

The correct match is

(1) a(iv), b(iii), c(i), d(ii)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iii), b(iv), c(ii), d(i)

73. The correct order of covalent nature is

(1) $NF_3 > PF_3 > BiF_3 > SbF_3$

(2) $NF_3 > BiF_3 > PF_3 > SbF_3$

(3) $NF_3 > PF_3 > SbF_3 > BiF_3$

(4) $NF_3 > SbF_3 > BiF_3 > PF_3$

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74. Given below are two statements:

Statement I: Ti^{2+} and V^{2+} liberate hydrogen gas from a dilute acid.

Statement II: The correct order of increasing oxidising power is $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$.

In the light of above statements, choose the correct answer from the options given below

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

75. Fusion of chromite ore with sodium carbonate in free access of air results in compound (X), Fe_2O_3 and a gaseous product CO_2. (X) is acidified with H_2SO_4 to give a coloured compound (Y). The products (X) and (Y) respectively are

(1) $NaCrO_2$ and Na_2CrO_4

(2) Na_2CrO_4 and $Na_2Cr_2O_7$

(3) $Na_2Cr_2O_7$ and Na_2CrO_4

(4) $NaCrO_2$ and $Na_2Cr_2O_7$

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76. Consider the following statements:

(a) White phosphorous is a translucent white waxy solid.

(b) White phosphorus dissolves in boiling NaOH solution in an inert atmosphere giving PH_3 and Na_2HPO_3

(c) Red phosphorus is insoluble in water as well as in carbon disulphide

(d) Red phosphorus glows in dark

The incorrect statements are

(1) (a) and (c) only

(2) (a), (b) and (c) only

(3) (b) and (d) only

(4) (b), (c) and (d) only

77. Identify the correct orders against the property mentioned

(a) $H_2Te > H_2O > H_2S$(Melting point)

(b) $H_2Se > H_2S > H_2O$(Dissociation constant in aqueous medium)

(c) S > Se > Te (Negative electron gain enthalpy)

(d) Te > Po > Se (Melting point)

Choose the correct answer from the options given below:

(1) (a), (b) and (c) only

(2) (b), (c) and (d) only

(3) (b) and (c) only

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(4) (a), (b), (c) and (d)

78. Which of the following statements are correct about Lanthanoids?

a. Both La^{3+} and Lu^{3+} are colourless

b. Mischmetal consists of 95% lanthanoid metal and 5% iron

c. Pr^{3+} and Yb^{2+} are paramagnetic in nature

d. Lanthanoids form $M(OH)_3$ hydroxide which are basic in nature (M = Lanthanoid elements)

The correct option is

(1) a, b and c only

(2) a, b and d only

(3) a, b, c and d

(4) b, c and d only

79. Which among the following is not a ambidentate ligand?

(1) SCN^-

(2) CN^-

(3) H_2O

(4) NO_2^-

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80. Orthophosphorous acid on heating gives

(1) H_3PO_2 and $H_4P_2O_7$

(2) PH_3 and H_3PO_4

(3) H_3PO_2 and H_3PO_4

(4) PH_3 and $H_4P_2O_6$

81. Which of the following compounds give cannizzaro reaction?

(a) 2-Methylpropanal

(b) Benzaldehyde

(c) Acetophenone

(d) 2,2-Dimethylpropanal

The correct option is

(1) (a), (c) and (d) only

(2) (a), (b) and (d) only

(3) (a) and (b) only

(4) (b) and (c) only

82. Correct order of reactivity of the following carboxylic acid towards esterification reaction is

(1) $HCOOH < CH_3COOH < C_2H_5COOH$

(2) $CH_3COOH < HCOOH < C_2H_5COOH$

CC-473 (3) $C_2H_5COOH < HCOOH < CH_3COOH$ CC-473

(4) $C_2H_5COOH < CH_3COOH < HCOOH$

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83. Consider the given reaction sequence

$$PhCH_2COOH \xrightarrow[(ii)\ \Delta]{(i)\ NH_3} A \xrightarrow[\Delta]{P_2O_5} B \xrightarrow[(ii)\ H_2O]{(i)\ DIBAL\text{-}H} C\ (Major)$$

Major product C is

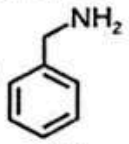
(1) $PhCH_2CH_2NH_2$

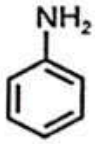
(2) $PhCH_2CHO$

(3) $PhCH_2CH_2OH$

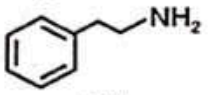
(4) $PhCH_2CH_3$

84. Which of the following compounds cannot be prepared by using Gabriel phthalimide synthesis?

(1) 

(2) 

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(3) 

(4) 

85. Salt reacts with warm dilute H_2SO_4, gives rotten egg like smell, indicates the presence of

(1) Cl^-

(2) CO_3^{2-}

(3) S^{2-}

(4) NO_3^-

86. All of the following are white precipitates, **except**

(1) $CaCO_3$

(2) $Mg(NH_4)PO_4$

(3) $PbSO_4$

(4) $BaCrO_4$

87. Lead chromate is soluble in hot sodium hydroxide solution due to the formation of

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(1) $Na_2[Pb(OH)_4]$

(2) $Na_4[Pb(OH)_2]$

(3) $Na_2[Pb(CrO_4)]$

(4) $Na_2[Pb(CrO_4)_2]$

88. Consider the following reaction schemes

Scheme 1: 
(P)

(i) Br_2(excess), H_2O
(ii) $NaNO_2$ + HCl, 273 K
(iii) CuCN/KCN
(iv) H_3O^+, Δ
(v) $SOCl_2$, Pyridine

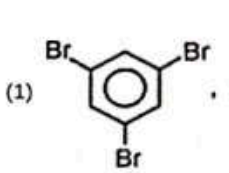
→ Q (Major)

Scheme 2: 
(R)

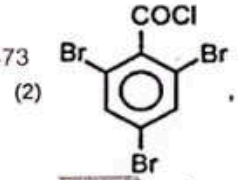
(i) Oleum
(ii) NaOH, Δ
(iii) H_3O^+
(iv) Br_2, CS_2, 273 K

→ S (Major)

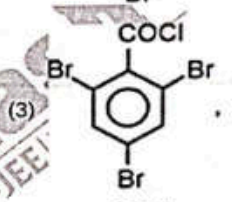
The products (Q) and (S) respectively are

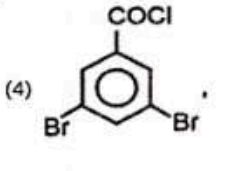
(1) 

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(2) 

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(3) 

(4) 

89. Given below are two statements. One is labelled as assertion (A) and other is labelled as reason (R)

Assertion (A) : 2, 4, 6 – trinitro –N, N-dimethylaniline is a stronger base than 2, 4, 6-trinitroaniline.

Reason (R) : The lone pair of electron on nitrogen in $-NMe_2$ group of 2, 4, 6- trinitro-N, N-dimethylaniline experiences inhibition of resonance due to bulky $-NO_2$ groups present at ortho positions.

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In the light of above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

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90. Consider the following statements

(a) The correct order of ionic radii is $K^+ > Li^+ > Al^{3+} > Mg^{2+}$
(b) Second ionisation enthalpy of carbon atom is less than that of boron atom.
(c) Negative electron gain enthalpy of element with atomic number 53 is higher than that of element with atomic number 16.
(d) The correct order of electronegativity (on Pauling scale) is B > Al > Mg > Be
(e) The correct order of acidic strength is $P_4O_{10} > Cl_2O_7 > SO_2$.
(f) The correct order of non-metallic character is F > Cl > S > O

The **incorrect** statements are

(1) (b), (d) and (f) only
(2) (c), (d) and (e) only
(3) (a), (d), (e) and (f) only
(4) (a), (c) and (e) only

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91. Which of the following taxa belong to the same rank?

(1) *Homo*, *Felis*, *pardus*
(2) Solanaceae, *leo*, Convolvulaceae
(3) *Mangifera*, *Solanum*, *Panthera*
(4) Hominidae, *Triticum*, Dicotyledonae

92. Lichen is a symbiotic association between

(1) Algae and roots of higher plants
(2) Algae and phycobiont
(3) Fungi and phycobiont
(4) Photosynthetic and chemoautotrophic organisms

93. *Paramoecium* and *Plasmodium*

(1) Both belong to the same kingdom as that of *Penicillium*
(2) Both have cilia as locomotory organs
(3) Both are endoparasites
(4) Both are unicellular protozoans

94. Members of deuteromycetes

(1) Have sex organs but do not reproduce sexually
(2) Help in mineral recycling
(3) Do not reproduce asexually
(4) Are all parasites

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95. Select the **incorrect** statement w.r.t. bryophytes.

(1) The plant body is attached to the substratum by unicellular or multicellular rhizoids
(2) They may possess stem-like and leaf-like structures
(3) The sporophyte is always free-living and photosynthetic
(4) Zygote does not undergo reduction division immediately

96. How many of the following algae are belonging to Brown algae and green algae respectively ?
Dictyota, *Volvox*, *Ectocarpus*, *Laminaria*, *Ulothrix*, *Spirogyra*, *Fucus*, *Chara*, *Porphyra*, *Polysiphonia*

(1) 4, 5
(2) 4, 4
(3) 5, 4
(4) 6, 3

97. Leaves of conifers do **not** have

(1) Needle like shape
(2) Very thin cuticle
(3) Sunken stomata
(4) Ability to withstand extremes of temperature

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98. Match the Column I with Column II and choose the **correct** option.

	Column I		Column II
A.	Pulvinus	(i)	In monocots, it expands into a sheath covering the stem partially or wholly.
B.	Leaf base	(ii)	It is a green expanded part of the leaf with veins and veinlets
C.	Petiole	(iii)	It is the swollen area in some leguminous plant where leaf is attached to the stem
D.	Leaf blade	(iv)	It allows the lamina to flutter in the wind

(1) A-(ii), B-(iii), C-(iv), D-(i)

(2) A-(iii), B-(i), C-(iv), D-(ii)

(3) A-(iii), B-(iv), C-(i), D-(ii)

(4) A-(iv), B-(iii), C-(i), D-(ii)

99. Racemose inflorescence differs from cymose inflorescence as in the former

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(1) The main axis terminates into a flower

(2) The peduncle has a limited growth

(3) Younger flowers are towards the apex and the older flower are present at the base

(4) Older flowers are larger than the younger flowers

100. Free central and basal type of placentation are observed respectively in

(1) Lemon and Mustard

(2) Marigold and *Primrose*

(3) *Dianthus* and Sunflower

(4) Mustard and Tomato

101. In flowering plants, pericycle

a. Gives rise to the lateral roots
b. Is made up of thin walled sclerenchymatous cells only
c. May act as storage organ of food materials
d. Is present in monocot stems
e. Gives rise to part of vascular cambium in monocot roots

The **correct** ones are

(1) a and c only

(2) d and e only

(3) a, b and c only

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(4) b, c and d only

102. Guard cells in grass leaf are

(1) Rectangular

(2) Barrel shaped

(3) Kidney shaped

(4) Dumb-bell shaped

103. Given below are some statements about the components of endomembrane system.

P. Golgi apparatus is extensive in cells specialised for secretion.
Q. Products of ER are modified during their transit from convex to concave face of Golgi apparatus.
R. The reticular structures of endomembrane system remain in close association.
S. The Golgi apparatus principally performs the function of modifying and packaging materials in entirely different but interconnected cis and convex faces, to be delivered to inter and intra-cellular targets.

(1) P and S are true and P is the reason for S

(2) P and R are true and P is the reason for R

(3) Q and R are true and Q is the reason for R

(4) Q and S are true and S is the reason for Q

104. The nuclear pores are complex structure which are formed by the

(1) Disintegration of nuclear membrane

(2) Fusion of outer and inner membranes of nucleus

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(3) Enzymatic action of lysosome

(4) Movement of RNA and protein molecules

105. Which of the following cell organelles is absent in higher plant cells but present in animal cells?

(1) Mitochondria

(2) Centrioles

(3) ER

(4) Plastids

106. Multi-layered epithelium that has limited role in secretion and absorption but provides protection against chemical and mechanical stresses is **not** found in

(1) Dry surface of skin

(2) Moist surface of pharynx

(3) Inner lining of pancreatic duct

(4) Inner lining of oviduct

107. Choose the **incorrect** statement.

(1) Each skeletal muscle fibre is composed of many long, cylindrical fibres arranged parallely.

(2) In some muscles, communication junctions at some fusion points allow the cells to contract as a unit.

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(3) In a specialised connective tissue the concentric rings in pliable matrix surround a central canal

(4) Long bones of the legs serve weight bearing functions.

108. Choose the **incorrect** match.

(1) Toxins – Abrin, Ricin

(2) Drugs – Vinblastine, Curcumin

(3) Essential oil – Concanavalin A

(4) Alkaloids – Morphine, Codeine

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109. Read the given statements and state them as true (T) or false (F).
a. Antibody does not fight against infectious agents.
b. GLUT-4 enables glucose transport into cells.
c. Trypsin is an active proteolytic enzyme.
Choose the suitable option.

	a	b	c
(1)	F	F	F
(2)	T	F	F
(3)	F	T	T
(4)	T	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

110. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Collagen	(i)	Most abundant protein in the whole of the biosphere
b.	RuBisCO	(ii)	Most abundant protein in animal world
c.	Cellulose	(iii)	Hormone that increases blood glucose levels
d.	Glucagon	(iv)	Homopolymer which is also a secondary metabolite

(1) a(ii), b(iv), c(iii), d(i)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(ii), b(i), c(iv), d(iii)

111. What is common between DNA, GTP and NAD?

(1) All of them are polymers of nucleotides

(2) Presence of pentose sugar

(3) Presence of adenine, as the nitrogenous base

(4) All of them possess the same kind of pyrimidines

112. Read the following statements w.r.t humans:
Statement (A): Breathing is a chemical process, while cellular respiration is purely mechanical.
Statement (B): Breathing ensures oxygen intake and carbon dioxide removal while cellular respiration is utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2.
Choose the **correct** option.

(1) Both the statements (A) and (B) are correct

(2) Only statement (A) is correct

(3) Both the statements (A) and (B) are incorrect

(4) Only statement (B) is correct

113. Choose the **correct** option w.r.t. residual volume.

(1) It allows complete emptying of lungs during forced expiration

(2) It can be measured by a simple spirometer

(3) It remains in the lungs even after forceful expiration

(4) It is not the component of FRC

114. Select the **correct** option w.r.t. first heart sound 'lub'.

(1) It is produced due to closure of atrio-ventricular valves during ventricular systole.

(2) It is produced due to closure of semilunar valves during ventricular diastole.

(3) It is produced due to closure of atrioventricular valves during ventricular diastole.

(4) It is produced due to opening of semilunar valves during atrial systole.

115. Which of the following is an **incorrect** match between column I and column II?

	Column I		Column II
A.	Parasympathetic neural signal	–	Decreases heart rate
B.	Cardiac arrest	–	When heart is not pumping blood effectively to meet needs of the body
C.	SA node	–	Pacemaker
D.	P-wave in ECG	–	Depolarisation of atria

(1) A

(2) B

(3) C

(4) D

116. Nucleolus is the site of synthesis of ribosomal RNA which are the components of sub- units

(1) 30S, 50S

(2) 60S, 50S

(3) 40S, 60S

(4) 30S, 40S

117. Arrange the different processes in **correct** order that occur in prophase I and choose the option accordingly.
I. Dissolution of synaptonemal complex
II. Formation of recombination nodule
III. Assembling of meiotic spindle at the poles
IV. Synapsis of homologous chromosomes

(1) IV, II, I, III

(2) II, I, IV, III

(3) III, I, IV, II

(4) IV, I, II, III

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118. Key event(s) of telophase is/are
(a) Reappearance of nucleolus, golgi complex and E.R.
(b) Chromosomes cluster at opposite poles of spindle apparatus.
(c) Chromosomes loss their identity as discrete elements

(1) Only (a) is incorrect

(2) Only (b) is correct

(3) Both (b) & (c) are incorrect

(4) All (a), (b) & (c) are correct

119. A diploid cell has 42 chromosomes in G_1 phase, then how many bivalents will be formed when the cell enters zygotene stage of prophase-I of meiosis?

(1) 42

(2) 21

(3) 63

(4) 84

120. Select the **incorrect** difference between C_3 pathway and C_4 pathway.

	C_3 Pathway	C_4 Pathway
(i)	Its first stable product is 3-phosphoglycerate	Its first stable product is oxaloacetic acid
(ii)	It is a slower process of carbon fixation	It is a faster process of carbon fixation
(iii)	Its primary acceptor of CO_2 is RuBP-a five carbon compound	Its primary acceptor of CO_2 is PEP - a three carbon compounds
(iv)	It occurs in the mesophyll and bundle-sheath cells of the leaves	It occurs in the mesophyll cells of the leaves

(1) (i)

(2) (iv)

(3) (iii)

(4) (ii)

121. The reaction centre chlorophyll a molecule in PS – II has an absorption maxima at

(1) 700 nm

(2) 400 nm

(3) 680 nm

(4) 200 nm

122. How many ATP molecules will be produced from the complete oxidation of one molecule of acetyl CoA?

(1) 12

(2) 14

(3) 36

(4) 38

123. How many molecules of CO_2 are released during glycolysis of two molecules of glucose?

(1) 6

(2) 12

(3) 3

(4) Zero

124. Initial surface areas of four leaves A, B, C and D were 5, 15, 10 and 20 cm^2. After one day their surface areas become 10, 20, 15 and 22 cm^2 respectively. Which leaf showed maximum relative growth rate?

(1) B

(2) A

(3) D

(4) C

125. Read the statements given below and choose the **correct** option.
(A) 2, 4-D is widely used as weedicide.
(B) Auxin controls the xylem differentiation.

(1) Both (A) and (B) are correct

(2) Only (A) is correct

(3) Only (B) is correct

(4) Both (A) and (B) are incorrect

126. Read the following statements and select the **correct** ones for pollen grains.
(a) About 25-50 µm in diameter
(b) Surrounded by a single layered wall
(c) Intine has prominent germ pores due to absence of cellulose
(d) Are well preserved due to presence of sporopollenin

(1) a & d only

(2) a, b & d

(3) a, c & d

(4) c & d only

127. Read the following statements to state them as True (**T**) or false(**F**) and select the **correct** option.
(A) A typical anther is bilobed with each lobe having two theca.
(B) The distal end of the filament is attached to thalamus or the petal of the flower.
(C) The process of formation of microspores from pollen mother cell through mitosis is called microsporogenesis.
(D) Carrot grass that came into India as contaminant with imported wheat, has become ubiquitous in occurrence and cause pollen allergy.

	(A)	(B)	(C)	(D)
(1)	T	T	F	T
(2)	T	F	T	F
(3)	T	F	F	T
(4)	F	F	T	T

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128. Both autogamy and geitonogamy are prevented in

(1) Maize

(2) Wheat

(3) Pea

(4) Papaya

129. Read the following statements.
(a) Code for a pair of contrasting traits
(b) They are slightly different forms of the same gene.
(c) True breeding line for a trait will have either both alleles dominant or both alleles recessive
(d) Both the alleles in a zygote are always contributed by one parent
Which ones are **true** for alleles?

(1) Only (a) & (d)

(2) Only (a), (b) & (c)

(3) Only (c) & (d)

(4) Only (b) & (d)

130. Phenylketonuria is an inborn error of metabolism in which

(1) Phenylalanine is converted to tyrosine

(2) Tyrosine is converted to phenylalanine

(3) Affected individual lacks phenylalanine hydroxylase enzyme

(4) Tyrosine get accumulated in brain resulting in mental retardation

131. Choose the incorrect statement regarding mechanism of urine formation in humans.

(1) Tubular secretion takes place in PCT.

(2) Aldosterone induces more reabsorption of Na^+.

(3) The counter current system contributes in diluting urine.

(4) Glomerular filtration rate is about 125 mL/min in an adult healthy man.

132. Which of the following is an **incorrect** match regarding organism, their mode of excretion and their respective structures involved significantly in removal of nitrogenous excretory waste?

	Organism	Mode of excretion	Structure involve in nitrogenous waste removal
(1)	Cockroaches	Uricotelism	Malphighian tubules
(2)	Prawns	Ammonotelism	Green glands
(3)	Bony fishes	Ureotelism	Kidneys
(4)	Human	Urotelism	Kidneys

(1) (1)

(2) (2)

(3) (3)

(4) (4)

133. Select the **incorrect** statement w.r.t. cortical nephrons.

(1) Loop of Henle is short and extends superficially in the medulla

(2) They are the most abundant type of nephrons in kidneys

(3) Vasa recta is well developed in cortical nephrons

(4) Malpighian corpuscle, PCT and DCT of these nephrons are situated in outer cortex

134. On the basis of following characters, identify 'A' and 'B' in an adult human.
A. Its cells are called chondrocytes.
B. Presence of hard matrix due to calcium salts.
Choose the correct option

(1) A-Cartilage present in the tip of nose, B-Sternum

(2) A-Frontal bone, B-Sternum

(3) A-Hyaline cartilage, B-Fibrous cartilage

(4) A-Fibula, B-Pubic symphysis

135. Which of the following correctly depicts the order of events required for muscle contraction?
a. Impulse stimulates sarcoplasmic reticulum
b. Release of calcium ions
c. Muscle contraction
d. Nerve impulse reaches neuro-muscular junction
e. Cross-bridge formation

(1) a → d → b → e → c

(2) d → b → a → e → c

(3) d → a → b → c → e

(4) d → a → b → e → c

136. An auto-immune disorder that affects neuromuscular junction leading to paralysis of skeletal muscles is called

(1) Muscular dystrophy

(2) Rheumatoid arthritis

(3) Osteoporosis

(4) Myasthenia gravis

137. **Assertion** : Impulse transmission across an electrical synapse is always faster than that across a chemical synapse.
Reason : Membranes of pre- and post-synaptic neurons are in very close proximity and gap junctions facilitate quicker transmission.
In the light of above statements, choose the **correct** option.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

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138. The proteinaceous hormone which increases blood pressure as well as water reabsorption is

(1) Thyroxine

(2) ANF

(3) Angiotensin II

(4) Aldosterone

139. How many functions given in the box below are performed by glucocorticoids?
Lipolysis, piloerection, glycogenolysis, increase in utilization of amino acids, immunosupression, gluconeogenesis
Select the **correct** option.

(1) 3

(2) 4

(3) 2

(4) 6

140. Choose the **incorrect** feature w.r.t *Hydra*.

(1) It is considered as a fresh water polyp

(2) Hypostome opens into mouth through gastrovascular cavity

(3) It has tentacles for locomotion

(4) Cnidoblasts are present on tentacles and on the body

141. Mark the **correct** statement in context of all platyhelminthes.

(1) They have blind sac body plan

(2) They are metamerically segmented

(3) They have hooks and suckers that help in attachment and ingestion

(4) They have flame cells that help in excretion and osmoregulation

142. Select the option that represents the **correct** combination of aquatic gnathostomes that exhibit oviparity.

(1) *Scoliodon*, *Chelone*, *Balaenoptera*

(2) *Bufo*, *Hyla*, *Chameleon*

(3) *Exocoetus*, *Betta*, *Clarias*

(4) *Carcharodon*, *Catla*, *Macaca*

143. Comprehend the given statements.
Statement (A) : In some countries, the muscular legs of frogs are used as food by man.
Statement (B) : Frogs maintain ecological balance because they serve as an important link of food chain and food web in the ecosystem.
Select the **correct** option.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct.

(4) Only statement (B) is correct

144. Choose the **correct** set of structures in cockroach which are paired

(1) Labium, Maxilla, Mandible, Hypopharynx

(2) Antenna, Anal style, Anal cerci, Collaterial glands

(3) Ocelli, Seminal vesicle, Testis, Labium

(4) Tegmina, Labrum, Leg, Frons

145. Read the following statements:
(a) Capacitation of sperms occurs in female reproductive tract.
(b) Menstrual flow results due to breakdown of endometrial lining of uterus.
(c) Ectopic pregnancy is referred to as implantation of embryo at site other than the uterus.
(d) The amnion of mammalian embryo is derived from endoderm and mesoderm.
Choose the option which contains only **incorrect** statement(s).

(1) a and b

(2) d only

(3) b and c

(4) a, b and c

146. Select the **odd** one w.r.t. post-fertilisation events.

(1) Development of blastocyst

(2) Formation of placenta

(3) Transfer of sperms into female genital tract

(4) Delivery of the foetus

147. Match column I with column II and choose the option with all **correct** matches.

	Column I		Column II
a.	Insemination	(i)	Delivery of the baby
b.	Implantation	(ii)	Embryonic development
c.	Gestation	(iii)	Attachment of blastocyst to uterine wall
d.	Parturition	(iv)	Transfer of sperms into the female genital tract

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iii), b(iv), c(i), d(ii)

148. Calculate the total number of polar bodies produced in a female if she gained her puberty at age 17 and is now 32 years old with one child. The given female has a menstrual cycle of 30 days. She breastfed her newly born for 6 months. Consider her reproductive span as 16 years.

(1) 157

(2) 208

(3) 178

(4) 220

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149. Consider the statements given below w.r.t humans.
Statement A: Human emotions arise mainly from interactions between the cerebral cortex and the limbic system.
Statement B: The cerebellum integrates information received from the semicircular canals of the ear and the auditory system.
Choose the **correct** option.

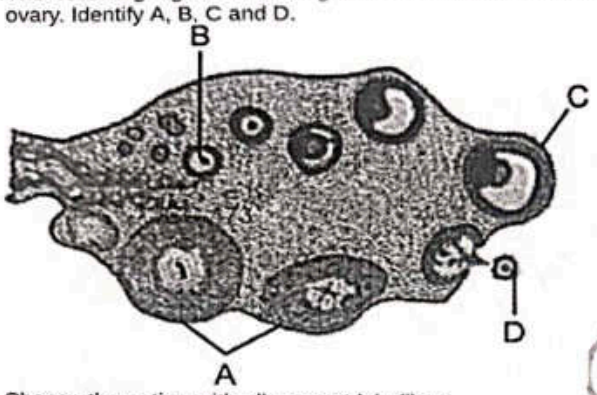
(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is incorrect

(4) Only statement B is incorrect

150. The following figure is a diagrammatic sectional view of ovary. Identify A, B, C and D.



Choose the option with all **correct** labellings.

	A	B	C	D
(1)	Graafian follicle	Ovum	Tertiary follicle	Corpus luteum
(2)	Corpus luteum	Primary follicle	Graafian follicle	Secondary oocyte
(3)	Ovum	Graafian follicle	Primary follicle	Corpus luteum
(4)	Primary follicle	Secondary follicle	Tertiary follicle	Corpus luteum

(1) (1)

(2) (2)

(3) (3)

(4) (4)

151. Read the following features w.r.t. surgical methods of contraception.
(a) This method is also called sterilisation.
(b) Considered as terminal method of contraception.
(c) Block gamete transport.
(d) Called vasectomy in female and tubectomy in males.
(e) High reversibility and contraceptive efficiency.
Choose the option, representing only correct statements.

(1) (a), (b), (c)

(2) (b), (c), (d)

(3) (c), (d), (e)

(4) (a), (d), (e)

152. Which of the following best describes the mechanism of action of Cu-T, when used as an emergency contraceptive?

(1) Copper ions inhibit the secretion of hCG from the trophoblast, leading to luteolysis

(2) Copper ions induce an inflammatory response in the endometrium, making it hostile to blastocyst implantation.

(3) Copper ions thicken the cervical mucus to prevent the zygote from entering the uterus.

(4) Cu-T physically blocks the fallopian tubes, preventing the zygote from descending into the uterus

153. The type of natural selection which gives two peaks form of graphical representation is

(1) Stabilising selection

(2) Directional selection

(3) Disruptive selection

(4) Balancing selection

154. In a population of 200 individuals, 98 exhibit recessive phenotype. What percentage of population is homozygous dominant?

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(1) 7%

(2) 9%

(3) 49%

(4) 42%

155. Colostrum is very essential to impart immunity to the newborn because it contains

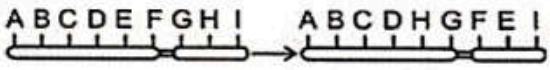
(1) Immunoglobulin A

(2) Macrophages

(3) Monocytes

(4) Natural killer cells

156. In the following figure a particular type of mutation is shown. Identify the mutation type.



(1) Deletion

(2) Duplication

(3) Translocation

(4) Inversion

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157. Sex of the chicks is determined by

(1) Their female parent

(2) Both of their male and female parents

(3) The temperature at which the egg is incubated

(4) Their male parent

158. A bacterial DNA having ^{14}N is allowed to replicate for one generation in medium containing $^{15}NH_4Cl$ as the only source of nitrogen and then for second generation in medium containing $^{14}NH_4Cl$. What will be the percentage of light, heavy and hybrid DNA respectively, obtained in second generation?

(1) 25%, 25%, 50%

(2) 50%, 50%, 0%

(3) 50%, 25%, 25%

(4) 50%, 0%, 50%

159. 'UGA does not code for any amino acid'. All of the below given conclusions can be withdrawn from the above statement, **except**

(1) UGA is a nonsense codon

(2) There is no tRNA for the codon UGA

(3) Genetic code is degenerate

(4) Translation terminates when ribosome reaches UGA of mRNA

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160. Satellite DNA

(1) Codes for proteins which are needed in the chloroplast

(2) Codes enzymes for Krebs' cycle

(3) Does not code for proteins

(4) Form a little portion of human genome

161. How many amino acids can be translated from the given sequence of nucleotides in mRNA if 8th and 11th nucleotide undergo deletion?
AUGUGGUCGCAUAAU

(1) 5

(2) 4

(3) 2

(4) 3

162. What is/are the contribution(s) of Avery, Macleod and McCarty?
(i) They concluded that DNA is the hereditary material.
(ii) They discovered that proteases and RNases affect transformation
(iii) They determined the biochemical nature of transforming principle.
Choose the option for **correct** one(s).

(1) (i), (ii) and (iii)

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(2) Only (i) and (iii)

(3) Only (ii) and (iii)

(4) Only (i)

163. Which of the given is used as immunosuppressive agent in organ-transplant patients?

(1) Statin

(2) Lipase

(3) Cyclosporin A

(4) Streptokinase

164. Which of the following is/are biofertilisers?
(a) Lactic acid bacteria
(b) *Anabaena*
(c) *Azotobacter*
(d) *Nostoc*

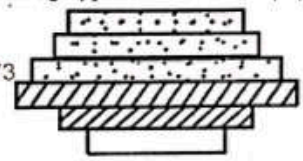
(1) Only (a)

(2) Only (a) and (b)

(3) Only (a) and (d)

(4) All, except (a)

165. An age pyramid for human population is shown below.



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Which of the following conclusion can be drawn from the above given age pyramid?

(1) It represents that the population is young or growing

(2) It is a pyramid of mature population

(3) It shows that population is stable

(4) It shows that population size is declining with time.

166. Read the following assertion and reason and choose the **correct** option.
Assertion : The Mediterranean orchid *Ophrys* employs sexual deceit' for pollination by a species of bee.
Reason : All orchids offer rewards to get pollination done.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

167. Read the following statements and select the **correct** option.
Statement A : Gross primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis.
Statement B : Net primary productivity is the available biomass for the consumption to herbivores and decomposers.

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(1) Only statement **A** is correct

(2) Only statement **B** is correct

(3) Both the statements **A** and **B** are correct

(4) Both the statements **A** and **B** are incorrect

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168. Which is not key functional aspect of ecosystem?

(1) Nutrient cycling

(2) Energy flow

(3) Productivity

(4) Stratification

169. Tropical regions have greater biological diversity than temperate ones. It is due to

(1) Relative stability of tropical regions

(2) Higher availability of solar energy in temperate regions

(3) Less availability of time for species diversification in tropical regions

(4) Negligible occurrence of glaciations in temperate regions

170. Choose the **odd** one for *ex-situ* conservation strategy?

(1) Seed bank

(2) Botanical garden

(3) Wildlife safari park

(4) National park

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171. All of the given are characteristic features of malignant tumors **except**

(1) Loss of contact inhibition

(2) Can metastasize

(3) Confined to original location

(4) Rapidly growing

172. Read the following statements and choose the **correct** option.

Statement-A : Bacteria like *Streptococcus pneumoniae* and *Haemophilus influenzae* are responsible for causing pneumonia in humans.

Statement-B : *Streptococcus pneumoniae* and *Haemophilus influenzae* infect the alveoli of the lungs.

(1) Both statements are incorrect

(2) Only statement A is correct

(3) Only statement B is incorrect

(4) Both statements are correct

173. If a circular DNA possesses 6 restriction sites for *Bam*HI then how many fragments will be produced upon their digestion with same restriction enzyme?

(1) 6

(2) 7

(3) 8

(4) 9

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174. Choose the **incorrect** statement.

(1) DNA ligase helps in ligation of DNA fragments

(2) Endonucleases break internal phosphodiester bonds at palindromic sites that are highly specific.

(3) Exonuclease removes nucleotides from the end of a DNA molecule.

(4) *Sal* I was the first discovered restriction endonuclease

175. Downstream processing includes

(1) Separation and purification

(2) Continuous culture system

(3) Expression of desired product

(4) Biosynthesis in bioreactors

176. Which of the following statements is incorrect for restriction endonucleases ?

(1) They belong to larger class of enzymes called nucleases

(2) Endonucleases recognise a specific palindromic nucleotide sequences in the DNA

(3) Each restriction endonuclease functions by inspecting the length of a DNA sequence

(4) *Eco*RI enzyme cut each of the two strands of double helix at any random points not related to their recognition sequences

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177. Read statements A and B and select the **correct** option.

Statement-A : DNA is a hydrophilic molecule, thus it cannot pass through cell membrane.

Statement-B : Electroporation creates transient microscopic pores in cell membrane making entry of DNA molecules into the cell easier.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct but B is incorrect

(4) Statement A is incorrect but B is correct

178. Select the **incorrect** statement w.r.t. RNAi.

(1) It is a naturally occurring mechanism that leads to the silencing of certain genes

(2) This mechanism is used for restricting the translation of certain mRNAs

(3) This mechanism provides defence against pathogens

(4) It takes place in all prokaryotic organisms as cellular defence mechanism

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179. How many among the following statements is/are **correct**?
I. GEAC makes decisions regarding patents of various new discoveries in India.
II. About 95% of all existing transgenic animals are mice.
III. Among 200,000 varieties of rice in India, Basmati variety was patented by a US company.
IV. RNAi is a post-translational method of gene silencing.
Choose the correct option.

(1) Four

(2) Three

(3) Two

(4) One

180. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection and is based on antigen-antibody interaction?

(1) Hybridization technique

(2) Western blotting

(3) Southern blotting

(4) ELISA

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16/06/2026

CODE-P

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PROJECT PHOENIX FOR RE-NEET-2026_CST-IOP

MM: 720 **CST-IOP** **Time: 180 Mins.**

Answers

1.	(1)	37.	(1)	73.	(3)	109.	(3)	145.	(2)
2.	(2)	38.	(1)	74.	(1)	110.	(4)	146.	(3)
3.	(4)	39.	(2)	75.	(2)	111.	(2)	147.	(2)
4.	(2)	40.	(3)	76.	(3)	112.	(4)	148.	(3)
5.	(4)	41.	(2)	77.	(2)	113.	(3)	149.	(1)
6.	(3)	42.	(4)	78.	(2)	114.	(1)	150.	(2)
7.	(1)	43.	(3)	79.	(3)	115.	(2)	151.	(1)
8.	(3)	44.	(1)	80.	(2)	116.	(3)	152.	(2)
9.	(3)	45.	(2)	81.	(2)	117.	(1)	153.	(3)
10.	(3)	46.	(2)	82.	(4)	118.	(4)	154.	(2)
11.	(3)	47.	(2)	83.	(2)	119.	(2)	155.	(1)
12.	(2)	48.	(3)	84.	(2)	120.	(2)	156.	(4)
13.	(2)	49.	(1)	85.	(3)	121.	(3)	157.	(1)
14.	(4)	50.	(4)	86.	(4)	122.	(1)	158.	(4)
15.	(3)	51.	(1)	87.	(1)	123.	(4)	159.	(3)
16.	(2)	52.	(4)	88.	(3)	124.	(2)	160.	(3)
17.	(3)	53.	(2)	89.	(4)	125.	(1)	161.	(4)
18.	(3)	54.	(4)	90.	(3)	126.	(1)	162.	(2)
19.	(2)	55.	(3)	91.	(3)	127.	(3)	163.	(3)
20.	(4)	56.	(4)	92.	(3)	128.	(4)	164.	(4)
21.	(3)	57.	(2)	93.	(4)	129.	(2)	165.	(4)
22.	(3)	58.	(3)	94.	(2)	130.	(3)	166.	(3)
23.	(1)	59.	(4)	95.	(3)	131.	(3)	167.	(3)
24.	(2)	60.	(3)	96.	(2)	132.	(3)	168.	(4)
25.	(4)	61.	(2)	97.	(2)	133.	(3)	169.	(1)
26.	(2)	62.	(2)	98.	(2)	134.	(1)	170.	(4)
27.	(4)	63.	(2)	99.	(3)	135.	(4)	171.	(3)
28.	(4)	64.	(2)	100.	(3)	136.	(4)	172.	(4)
29.	(4)	65.	(4)	101.	(1)	137.	(1)	173.	(1)
30.	(1)	66.	(1)	102.	(4)	138.	(3)	174.	(4)
31.	(3)	67.	(3)	103.	(3)	139.	(1)	175.	(1)
32.	(3)	68.	(4)	104.	(2)	140.	(2)	176.	(4)
33.	(1)	69.	(4)	105.	(2)	141.	(4)	177.	(1)
34.	(4)	70.	(1)	106.	(4)	142.	(3)	178.	(4)
35.	(3)	71.	(3)	107.	(3)	143.	(1)	179.	(3)
36.	(1)	72.	(4)	108.	(3)	144.	(2)	180.	(4)

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Hints and Solutions

PHYSICS

(1) **Answer :** (1)
Solution:
Difference of f_A and f_B is 6 Hz
If tension decreases, f_B decreases and becomes f_B'.
Now, difference of f_A and f_B' = 7 Hz (increases)
So, $f_A > f_B$
$f_A - f_B = 6$ Hz
$f_A = 530$ Hz
$f_B = 524$ Hz (original)

(2) **Answer :** (2)
Solution:
Power of one part:
Power $P = \frac{1}{f} = (\mu - 1)\left[\frac{1}{\infty} - \left(-\frac{1}{R}\right)\right] = \frac{\mu - 1}{R}$

(3) **Answer :** (4)
Hint:
$\vec{E}$ inside a conductor will always be zero.
Solution:
$\vec{E} = \vec{E}_Q + \vec{E}_{in}$
$\vec{E}_{in} = -\vec{E}_Q$ [$\because \vec{E} = 0$]
$\vec{E}_{in} = -\vec{E}_Q = \frac{1}{16\pi\varepsilon_0}\frac{Q}{R^2}$ towards left

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(4) **Answer :** (2)
Solution:
$R = 4H \cot(\theta)$
Given, $R = 8H$
$\Rightarrow 8H = 4H \cot(\theta)$
$\Rightarrow \cot(\theta) = 2$
$\theta = \cot^{-1}(2)$

(5) **Answer :** (4)
Solution:
For diamagnetic material, susceptibility is independent of temperature and it is negative.

(6) **Answer :** (3)
Solution:
$\therefore$ Current, $l = \frac{V}{R} = \frac{10}{2} = 5\,A$

(7) **Answer :** (1)
Solution:
We know, $\tau = I\vec{A} \times \vec{B}$
$\tau = I\,A\,B\,\sin\theta$
Here $\theta = 0°$
$\boxed{\therefore \tau = 0}$

(8) **Answer :** (3)
Solution:
$\Delta\vec{x} = 2\hat{i}$
$\Delta\vec{y} = -3\hat{j}$

$\left|\vec{v}_{av}\right| = \frac{\Delta s}{\Delta t} = \frac{\sqrt{4+9}}{1} = \sqrt{13}$ unit

(9) **Answer :** (3)
Solution:

$V_T = \frac{2(\rho-\sigma)gr^2}{9\eta}$

$V_T \propto r^2$

$\frac{V_0}{4V_0} = \frac{r^2}{R^2}$

R = 2r

$\frac{M}{m} = \frac{\left(\frac{4}{3}\pi R^3\right)\rho}{\left(\frac{4}{3}\pi r^3\right)\rho} = 8$

(10) **Answer :** (3)
Solution:

For third overtone $4 \times \frac{\lambda}{2} = I \Rightarrow \lambda = \frac{I}{2}$

for $x = \frac{I}{3}, kx = \frac{2\pi}{\lambda}\frac{I}{3} \Rightarrow kx = \frac{4\pi}{3}$

y = 2Asinkx here 2A = a(maximum amplitude of wave)

$y = a\sin\frac{4\pi}{3} \Rightarrow -a\sin\frac{10}{3} \Rightarrow -\frac{\sqrt{3}a}{2}$

(11) **Answer :** (3)
Solution:

5 N = 5a

$\Rightarrow a$ = 1 m/s^2

Acceleration of pully = $\frac{a}{2}$

(12) **Answer :** (2)
Solution:

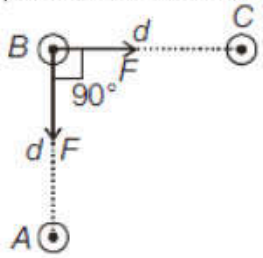
Concepts of conversion of Galvanometer into Ammeter and Voltmeter.

(13) **Answer :** (2)
Solution:

At resonance, power factor (cos ϕ) = 1

(14) **Answer :** (4)
Solution:

Force between BC and AB will be same in magnitude.



$F_{BC} = F_{BA} = \frac{\mu_0 I^2}{2\pi d}$

$F = \sqrt{2}F_{BC}$

$= \sqrt{2}\frac{\mu_0}{2\pi}\frac{I^2}{d}$

$F = \frac{\mu_0 I^2}{\sqrt{2}\pi d}$

(15) **Answer :** (3)
Solution:

Equilibrium position of the neutral point from mass 'm' is $= \left(\frac{\sqrt{m}}{\sqrt{m}+\sqrt{M}}\right)d$

$V_1 = \frac{-Gm_1}{r_1};\ V_2 = \frac{-Gm_2}{r_2}$

$V_1 = \frac{-Gm}{\sqrt{m}d}\left(\sqrt{M}+\sqrt{m}\right);\ V_2 = \frac{-GM}{\sqrt{M}d}\left(\sqrt{M}+\sqrt{m}\right)$

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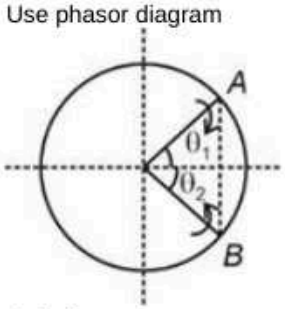
$V_1 = \frac{-G}{d}\sqrt{m}\left(\sqrt{M}+\sqrt{m}\right)$; $V_2 = \frac{-G}{d}\sqrt{M}\left(\sqrt{M}+\sqrt{m}\right)$

$V = \frac{-G}{d}(\sqrt{M}+\sqrt{m})^2$

(16) Answer : (2)

Hint:

Use phasor diagram



Solution:

$A\cos\theta_1 = \frac{A}{2} \Rightarrow \theta_1 = \frac{\pi}{3}$

$A\ \cos\theta_2 = \frac{A}{2} \Rightarrow \theta_2 = \frac{\pi}{3}$

Phase difference

$= \theta_1 + \theta_2$

$= \frac{2\pi}{3}$

(17) Answer : (3)

Solution:

Binding energy per nucleon is maximum for C^{12} then for He^4 and least for Li^7. Hence C^{12} nucleus is most stable then He^4 and least is Li^7.

(18) Answer : (3)

Solution:

$x = ML^2T^{-2}$

$\frac{\Delta x}{x} \times 100 = 1 \times \frac{\Delta M}{M} \times 100 + 2\frac{\Delta L}{L} \times 100 + 2\frac{\Delta T}{T} \times 100$

% x = 1(2%) + 2(1%) + 2(1%)

= 6%

(19) Answer : (2)

Solution:

$u = \frac{1}{2}\varepsilon_0 E^2$

$= \frac{1}{2} \times 8.85 \times 10^{-12} \times 36$

$= 1.6 \times 10^{-10}$ J/m^3

(20) Answer : (4)

Solution:

The amplitude will be $A\ \cos 60^\circ = \frac{A}{2}$

(21) Answer : (3)

Hint:

Use appropriate formulae.

Solution:

(Quantity)	(Corresponding formula)
A. Torque τ	II. $I\alpha$
B. Angular momentum	III. $Mvr_\perp$
C. $\sum \vec{F}_{ext}$	I. $M\vec{a}_{cm}$
D. Instantaneous power	IV. $\vec{\tau}\cdot\vec{\omega}$

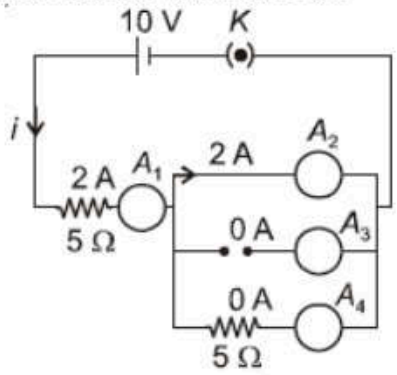
(22) Answer : (3)

Hint:

Inductor behaves like short circuit and capacitor behaves like open circuit at $t = \infty$

Solution:

After long time, circuit will be like



$i = \frac{10}{5} = 2\ A$

Hence reading of A_1 and A_2 = 2 A

Reading of A_3 and A_4 = 0

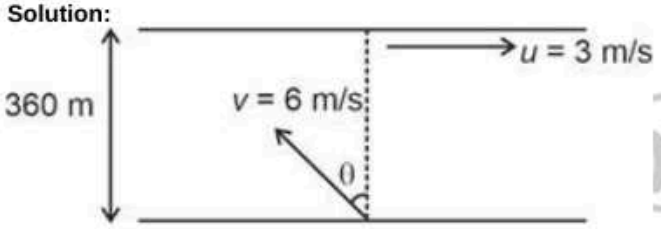
Hence A_1, A_2, A_3 and A_4 → 2 A, 2 A, 0, 0

(23) **Answer :** (1)

Hint:

For t_{min}, boat should be steered perpendicularly to the flow of river

Solution:



To cross the river in minimum time, θ = 0°

$t_{\min} = \frac{d}{v} = \frac{360}{6} = 60$ s

Drift = 3 × 60 = 180 m

(24) **Answer :** (2)

Hint:

In distinct vision mode, magnifying power of compound microscope

$M = \frac{v_0}{u_0}\left(1 + \frac{D}{f_e}\right)$ *and* $L = v_0 + u_e$

Solution:

For objective $\frac{1}{v_0} + \frac{1}{2.2} = \frac{1}{2}$

$\Rightarrow \frac{1}{v_0} = \frac{1}{2} - \frac{10}{22} = \frac{11-10}{22}$

$\Rightarrow v_0 = 22\ cm$

$M = \frac{v_0}{u_0}\left(1 + \frac{D}{f_e}\right) = \frac{22}{2.2}\left(1 + \frac{25}{5}\right)$

= 10 × 6 = 60

Distance of intermediate image from the eyepiece, when final image is at least distance of distinct vision,

$u_e = \frac{f_e D}{D + f_e} = \frac{5 \times 25}{5 + 25} = \frac{125}{30}$

$\approx 4\ cm$

$L = v_0 + u_e$

$\simeq 22 + 4$

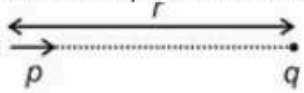
= 26 cm

(25) **Answer :** (4)

Solution: Answer (4)

Hint: Force on dipole due to charge is equal to force on charge due to dipole

Sol.:



$F_{\text{dipole}/q} = F_{q/\text{dipole}}$
$= q E_p$
$= q\frac{2kp}{r^3} = \frac{2kpq}{r^3}$

(26) **Answer :** (2)
Solution:
$d\varepsilon = \int_R^{2R} B\omega x\,dx$
$= \frac{B\omega}{2}\left[x^2\right]_R^{2R}$
$= \frac{3B\omega R^2}{2}$

(27) **Answer :** (4)
Solution:
(1) Point *A* is proportional limit. Till *A*, the stress is directly proportional to strain.
(2) Point *B* is yield point. After point *B*, if stress is increased there will be permanent extension even after removal of force.
(3) Point *C* is ultimate tensile strength. This is the maximum stress developed in given wire.
(4) Point *D* is fracture point.

(28) **Answer :** (4)
Solution:
If $\Delta U = 0$ $(\Delta Q = \Delta U + \Delta W)$
$\Delta Q = \Delta W$

(29) **Answer :** (4)
Hint:
Centripetal acceleration is given by $a_c = \frac{v^2}{r} = \omega^2 r$
Solution:
Here, $\omega = \frac{180\times 2\pi}{15\times 60} = \frac{24\pi}{60} = 0.4\pi$ rad/s
Hence, $a_c = \omega^2 r = (0.4\pi)^2 \times 6$
$= 0.16\pi^2 \times 6 \approx 9.6$ m s^{-2}
f = centripetal force = 100 × 9.6 = 960 N

(30) **Answer :** (1)
Solution:
$V_A = \frac{6}{9} \times 18 = 12$ V
$V_B = \frac{3}{9} \times 18 = 6$ V
$V_A - V_B = 6$ V
$Q = C_{\text{eq}}V = 4 \times 18\ \mu\text{C} = 72\ \mu\text{C}$

(31) **Answer :** (3)
Solution:
$y = a\sin\omega t$
$v = \frac{dy}{dt} = a\omega\cos\omega t = a\omega\cos\frac{2\pi}{T}\times\frac{T}{4} = a\omega\cos\frac{\pi}{2} = 0$

(32) **Answer :** (3)
Solution:
Since $L = n\frac{h}{2\pi}$, so for minimum value of L, n is taken as 1
$\Rightarrow\ L_{\min} = \frac{h}{2\pi}$

(33) **Answer :** (1)
Solution:
In a p-n junction diode, hole diffuse from p-region to n-region because there is large concentration of holes in p-region as compared to n-region.

(34) **Answer :** (4)
Solution:
Shift = 5 (fringe width)
$\frac{(\mu-1)tD}{d} = \frac{5\lambda D}{d}$

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$t = \frac{5\lambda}{\mu - 1} = 50000$ Å

(35) Answer : (3)
Solution:

$\tau = l\alpha = mg \times \left(\frac{I}{2}\cos 60^\circ\right)$

$\alpha = \frac{(1)\times 10\times\left(\frac{2}{2}\times\frac{1}{2}\right)}{\frac{ml^2}{3}} = \frac{5\times 3}{1\times(2)^2} = \frac{15}{4} = 3.75 \text{rad/s}^{-2}$

(36) Answer : (1)
Solution:

$\sin\theta = \frac{R}{d} = \frac{1}{\sqrt{2}}$

θ = 45°

$R = \frac{mv}{qB}$

After coming out *B* particle will collide with the wall
time to exit from field $t_1 = \frac{\pi m}{4qB}$

time spend in region of non magnetic field $t_2 = \frac{2m}{qB}$

time for returning journey in *B*, $t_3 = \frac{\pi m}{4qB}$

Total time taken $T = t_1 + t_2 + t_3$

$= \frac{\pi m}{4qB} + \frac{2m}{qB} + \frac{\pi m}{4qB}$

$= \frac{m}{2qB}[\pi + 4]$

(37) Answer : (1)
Solution:

Newtons law of cooling $\frac{\theta_1 - \theta_2}{\Delta t} = K\left[\frac{\theta_1 + \theta_2}{2} - \theta_0\right]$

First $\Rightarrow \frac{70-60}{5} = K[65 - \theta_0]$

$\Rightarrow 2 = K[65 - \theta_0]$ ...(i)

Next $\Rightarrow \frac{60-54}{5} = K[57 - \theta_0]$ ...(ii)

Diving (i) and (ii)

$\frac{5}{3} = \frac{65-\theta_0}{57-\theta_0}$

$\Rightarrow 285 - 5\theta_0 = 195 - 3\theta_0$

$\Rightarrow 2\theta_0 = 90$

$\theta_0 = 45^\circ$

(38) Answer : (1)
Hint:

Use, $\frac{C_P}{C_V} = \gamma$

Solution:

$\gamma = \frac{f+2}{f}$

Degree of freedom for monoatomic gas is 3 and for diatomic gas is 6.

$\gamma_{\text{mono}} = \frac{5}{3}$

$\gamma_{\text{dia}} = \frac{7}{5}$

(39) Answer : (2)
Hint:

$V_A = \frac{k\,p\cos\theta}{r^2}$

Solution:

$V_A = \frac{k\,p\cos 30^\circ}{r^2}$

$V_A = \frac{\sqrt{3}}{2}\frac{kp}{r^2}$

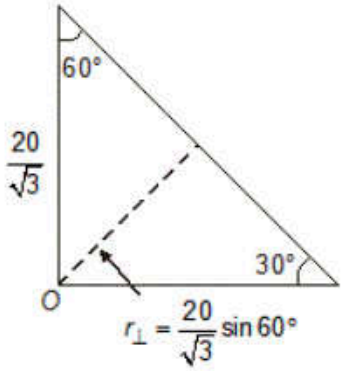
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(40) Answer : (3)
Solution:
$K \cdot E = h\nu - \phi$
K·E = (6·2 – 4·2) eV = 2 eV = $2 \times 1{\cdot}6 \times 10^{-19}$ J
$\boxed{K \cdot E = 3 \cdot 2 \times 10^{-19}\,\text{J}}$

(41) Answer : (2)
Solution:
Induced electric field is in anticlockwise direction, hence force on e^- is towards right.

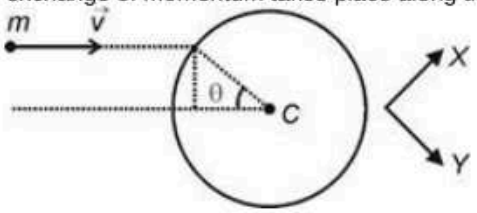
(42) Answer : (4)
Solution:
Adding detergent decreases the surface tension of water.

(43) Answer : (3)
Solution:



$L = mvr_\perp$
= 5 × (u + at) $\frac{20}{\sqrt{3}}\sin 60^\circ$
= 5 × [$g$sin 30° × 10]
= 250 kg m^2s^{-1}

(44) Answer : (1)
Solution:
Conserve the linear momentum of (particle +disc) along the normal. Since the mass of disc and particle are equal so exchange of momentum takes place along the normal.



$m\vec{v}_D = mv\cos\theta\hat{j}$
$\vec{v}_D = v\cos\theta\hat{j}$
From figure $\sin\theta = \frac{1}{2}$, θ = 30°
$\vec{v}_D = v\frac{\sqrt{3}}{2}\hat{j}$.

(45) Answer : (2)
Solution:
Energy of photon $E = \frac{hc}{\lambda} = h\nu$
and momentum, $P = \frac{h}{\lambda} = \frac{h\nu}{c}$

CHEMISTRY

(46) Answer : (2)

Solution:

$NaOH + HCl \rightarrow NaCl + H_2O$

1 mol 1 mol

$\Rightarrow$ (50 × 0.50)m mol of HCl will react with (50 × 0.50)m mol of NaOH

$\therefore$ mass of NaOH reacted = $50 \times 0.50 \times 10^{-3} \times 40$

= 1 g

$\therefore$ left amount of NaOH = 1g

(47) Answer : (2)

Solution:

$\Delta E = 13.6\ Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

For (6 → 5); $\Delta E = 13.6 \times 1^2 \left(\frac{1}{25} - \frac{1}{36}\right)$

For (5 → 3); $\Delta E = 13.6 \left(\frac{1}{9} - \frac{1}{25}\right)$

For (5 → 4); $\Delta E = 13.6 \left(\frac{1}{16} - \frac{1}{25}\right)$

For (6 → 1); $\Delta E = 13.6 \left(\frac{1}{1} - \frac{1}{36}\right)$

(48) Answer : (3)

Solution:

Na (Z = 11)

$1s^2\ 2s^2\ 2p^6\ 3s^1$

- For 1s, n + l = 1 + 0 = 1
- For 2s, n + l = 2 + 0 = 2, i.e., 1 orbital
- For 2p, n + l = 2 + 1 = 3
- For 3s, n + l = 3 + 0 = 3

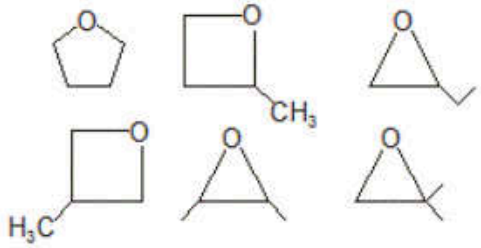
(49) Answer : (1)

Solution:

The number of degenerate orbitals in 3rd shell of H-atom is 9.

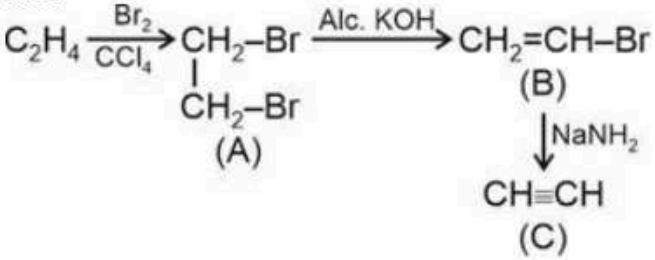
(50) Answer : (4)

Solution:



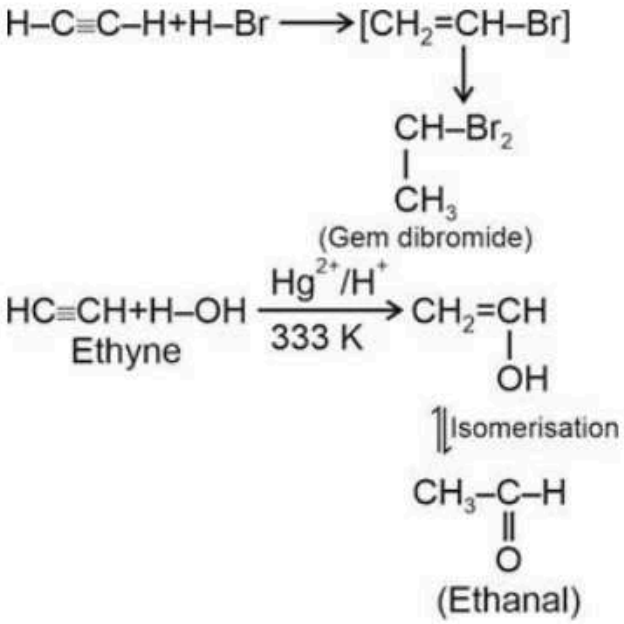
(51) Answer : (1)

Hint:



Solution:

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(52) **Answer :** (4)

Hint:

For isothermal reversible change

$w = -nRT \ln \frac{V_2}{V_1}$

Solution:

At constant temperature there is no change in internal energy.

ΔU = q + w

At constant temperature, ΔU = 0

$q = -w = nRT \ln \frac{V_2}{V_1}$

(53) **Answer :** (2)

Solution:

On balancing the given redox reaction:

$2\,MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2\,Mn^{2+} + 10\,CO_2 + 8H_2O$

5 mol of $C_2O_4^{2-} \equiv$2 mol of MnO_4^-

$\therefore$ 0.5 mol of $C_2O_4^{2-} \equiv$0.2 mol of MnO_4^-

(54) **Answer :** (4)

Solution:

$Al_2(SO_4)_3 \rightarrow \underset{2s}{2\,Al^{3+}} + \underset{3s}{3\,SO_4^{2-}}$

$K_{sp} = (Al^{3+})^2 (SO_4^{2-})^3$

$K_{sp} = (2s)^2 (3s)^3$

$K_{sp} = 4s^2 \times 27s^3$

$K_{sp} = 108s^5$

(55) **Answer :** (3)

Hint:

ΔG = ΔG° + RT ln Q

At equilibrium; ΔG = 0 and Q = K

Solution:

ΔG° = –RT × 2.303 log K

= –2 × 300 × 2.303 log 10

= –1.38 kcal mol^{-1}

(56) **Answer :** (4)

Hint:

Percentage of nitrogen = $\frac{28 \times \text{volume of } N_2 \text{ gas at STP} \times 100}{22400 \times \text{mass of organic compound}}$

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Solution:

$V_1 = 30$ mL

$P_1 = 715 - 15 = 700$ mm Hg

$T_1 = 300$ K

So, $V_2 = \frac{P_1 \times V_1 \times 273}{T_1 \times 760} = \frac{700 \times 30 \times 273}{300 \times 760}$

= 25.1 mL

Percentage of nitrogen = $\frac{28 \times 25.1 \times 100}{22400 \times 0.1}$

= 31.4%

(57) Answer : (2)

Solution:

Cyclic, planar species having (4n + 2)π electrons in conjugation are aromatic in nature.

(58) Answer : (3)

Hint:

Central atom of I_3^+ is sp^3 hybridised

Solution:

Species	Shape
XeF_4	Square planar
I_3^+	bent
PCl_3	Pyramidal
NO_2^+	Linear

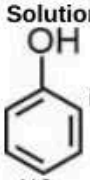
(59) Answer : (4)

Solution:

Rotation of carbon- carbon double bond is restricted.

(60) Answer : (3)

Solution:



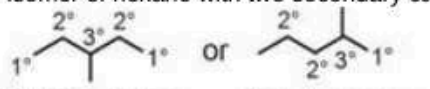
is most reactive towards electrophilic substation due to +R effect of –OH group on benzene ring.

$-NO_2$ and –COOH have –R effect while –Cl has –I and +R effect.

(61) Answer : (2)

Solution:

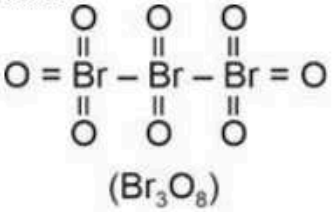
Isomer of hexane with two secondary carbon atoms and one tertiary carbon atom is



3-Methylpentane 2-Methylpentane

(62) Answer : (2)

Hint:



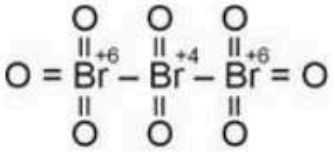
(Br_3O_8)

Solution:

- $\overset{-4+1}{C}H_4(g) + 2\overset{0}{O}_2 \xrightarrow{\Delta} \overset{+4-2}{C}O_2(g) + 2\overset{+1}{H}_2O^{-2}(l)$

is an example of combination redox reaction

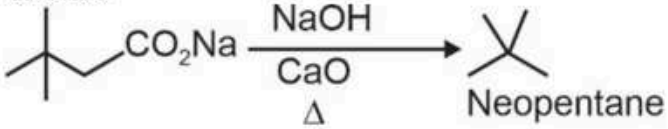
-



(Br_3O_8)

(63) Answer : (2)

Solution:



(64) Answer : (2)

Solution:

$\Delta S = 2.303 \ \ nR \ \log \frac{V_2}{V_1}$

$= 2.303 \times 1 \times 2 \log \frac{20}{2}$

$= 2.303 \times 2 \times \log 10 = 4.606 \ calK^{-1}mol^{-1}$

(65) Answer : (4)

Solution:

C → A; Reversible isothermal compression.

W = $-2.303nRT\log\frac{V_2}{V_1}$

= $-2.303 \times 3 \times 2 \times 300 \times \log\frac{30}{60}$

= $2.303 \times 1800 \times 0.3$

= 1.24 kcal

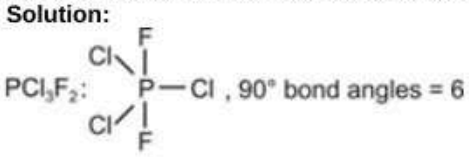
ΔH = 0, for cyclic process

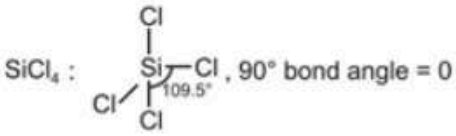
(66) Answer : (1)

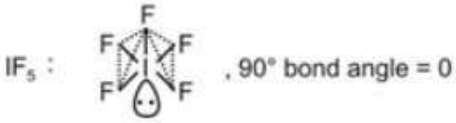
Hint:

Molecule having trigonal bipyramidal shape will have bond angle equal to 90°.

Solution:







(67) Answer : (3)

Solution:

Higher value of 'i' has lower freezing point.

(68) Answer : (4)

Solution:

Kohlrausch law states that limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of λ° of the anion and cation of electrolyte

$\Lambda_m^o = n\lambda^{o+} + n\lambda^{o-}$

Also, $\frac{\Lambda_m}{\Lambda_m^o} = \alpha$

And $K_a = \frac{C\alpha^2}{(1-\alpha)} = \frac{C\left(\frac{\Lambda_m}{\Lambda_m^o}\right)^2}{\left(1-\frac{\Lambda_m}{\Lambda_m^o}\right)}$

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(69) **Answer :** (4)
Solution:

$E_{H^+/H_2} = E^{\circ}_{H^+/H_2}$

$-\frac{0.0591}{2}\log\frac{P_{H_2}}{[H^+]^2}$

$\therefore E_{H^+/H_2} = 0 - \frac{0.0591}{2}\log\frac{1}{(10^{-2})^2}$

$= -\frac{0.0591}{2} \times 4 = -0.118\ V$

(70) **Answer :** (1)
Solution:
Rate of reaction will be increased by 27 times.

(71) **Answer :** (3)
Solution:
75% completion = 2 half lives

$6 = \frac{2 \times 0.693}{k}$

$\Rightarrow k = \frac{0.693}{3}$

(72) **Answer :** (4)
Hint:
Ionisation isomerism arises when the counter ion in a complex salt is itself a potential ligand and can displace a ligand which can then become the counter ion.
Solution:

Complex	Magnetic character	Isomerism
$[Cr(NH_3)_4Br_2]Br$	Cr^{3+} is d^3, hence paramagnetic	Cis-trans
$[Ti(H_2O)_5Br](NO_3)_2$	Ti^{3+} is d^1, hence paramagnetic	Ionization
$[Pt(en)(NH_3)Br]NO_3$	Pt^{2+} is d^8, complex is square planar, all electrons are paired	Ionization
$[Co(NH_3)_4(NO_3)_2]NO_3$	Co^{3+} is d^6, hence diamagnetic	Cis-trans

(73) **Answer :** (3)
Solution:
Down the group covalent character decreases.

(74) **Answer :** (1)
Solution:

$Ti^{2+}(aq) + 2H^+(aq) \rightarrow 2Ti^{3+}(aq) + H_2(g)$

The ions Ti^{2+} and V^{2+} are good reducing agents and will liberate H_2 gas from a dilute acid.

$M^{3+}\|M^{2+}$	E°(V)
Ti	–0.37
V	–0.26

(75) **Answer :** (2)
Solution:

$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow \underset{(X)}{8Na_2CrO_4} + 2Fe_2O_3 + 8CO_2$

$2Na_2CrO_4 + 2H^+ \rightarrow \underset{(Y)}{Na_2Cr_2O_7} + 2Na^+ + H_2O$

(76) **Answer :** (3)
Solution:
• White phosphorus dissolves in boiling NaOH solution in an inert atmosphere giving PH_3 and NaH_2PO_2
• $P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$
Red phosphorus does not glow in the dark.

(77) **Answer :** (2)
Solution:

Compounds	Melting point (K)
H_2Te	222

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H_2S	188
H_2O	273

(78) **Answer :** (2)
Hint:
Those compounds which contain unpaired e^- are paramagnetic in nature
Solution:
La^{3+} and Lu^{3+} have $4f^0$ and $4f^{14}$ electronic configuration respectively. So these are colourless in nature
Pr^{3+} has 2 unpaired electrons. So it is paramagnetic but Yb^{2+} has no unpaired electrons ($4f^{14}$) so it is diamagnetic in nature.

(79) **Answer :** (3)
Solution:
H_2O is a monodentate ligand.

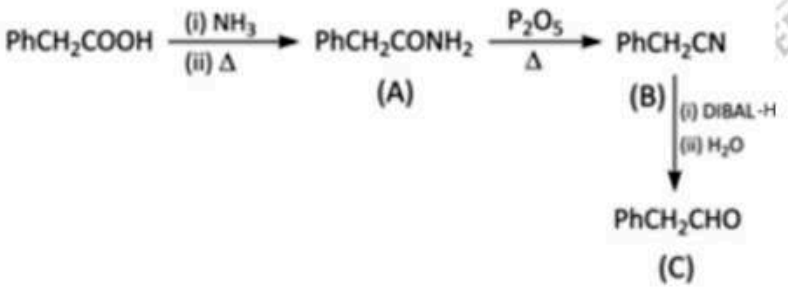
(80) **Answer :** (2)
Hint:
Orthophosphorous acid (H_3PO_3) undergoes disproportionation reaction on heating.
Solution:

$$4H_3PO_3 \xrightarrow{\Delta} 3H_3PO_4 + PH_3$$

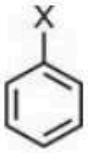
(81) **Answer :** (2)
Solution:
Aldehydes which do not contain an α-hydrogen atom, undergo self-oxidation and reduction (disproportionation) reaction on treatment with concentrated alkali.

(82) **Answer :** (4)
Solution:
Electrophilicity of carbon atom of carboxylic group decreases as the size of alkyl group attached to carboxylic group increases.

(83) **Answer :** (2)
Solution:



(84) **Answer :** (2)
Solution:

Aniline cannot be prepared by using Gabriel phthalimide as for this 

(aryl halide) is to be used and it cannot undergo

nucleophilic substitution.

(85) **Answer :** (3)
Hint:
H_2S gas has rotten egg like smell
Solution:

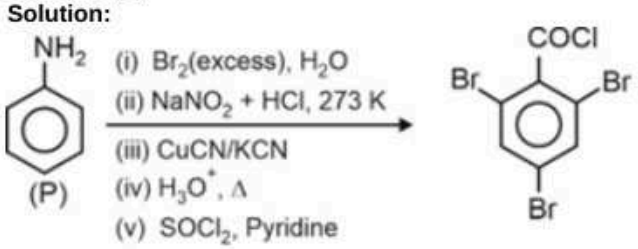
$$\underset{\text{(salt)}}{Na_2S} + H_2SO_4 \rightarrow Na_2SO_4 + H_2S\,(gas)$$

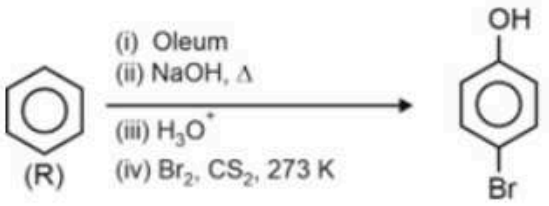
(86) **Answer :** (4)
Solution:
Barium chromate is a yellow coloured precipitate.

(87) **Answer :** (1)
Hint:
Lead chromate form sodium tetra hydroxoplumbate (II)
Solution:

$$PbCrO_4 + 4NaOH \rightleftharpoons \underset{\substack{\text{Sodiumtetra}\\\text{hydroxoplumbate(II)}}}{Na_2[Pb(OH)_4]} + Na_2CrO_4$$

(88) **Answer :** (3)
Solution:





(89) **Answer :** (4)
Solution:
Due to presume of bulky $-NO_2$ groups of ortho positions, conjugation of lone pair of $-NMe_2$ group with ring decrease due to which lone pair density on nitrogen is more due to which it become more basic.

(90) **Answer :** (3)
Solution:

Element	Electronegativity
Be	1.5
B	2.0
Mg	1.2
Al	1.5
N	3.0
O	3.5
P	2.1
S	2.5

• Electronegativity $\propto \frac{1}{\text{Metalic character}} \times (\text{Non metallic character})$

BIOLOGY

(91) **Answer :** (3)
Solution:
Mangifera, *Solanum*, *Panthera* all are genera.

(92) **Answer :** (3)
Solution:
Lichen is an association between algae (phycobiont) and fungi (mycobiont).

(93) **Answer :** (4)
Solution:
Paramoecium and *Plasmodium* both are unicellular.
Both belong to the group protozoa.

(94) **Answer :** (2)
Hint:

Members of deuteromycetes are good decomposers.
Solution:
Deuteromycetes help in mineral recycling.

(95) **Answer :** (3)
Solution:
In bryophytes, the sporophyte is not free-living but attached to the photosynthetic gametophyte and derives nourishment from it.

(96) **Answer :** (2)
Solution:
Brown alage – *Dictyota, Ectocarpus, Laminaria Fucus*
Green algae – *Volvox, Ulothrix, Spirogyra, Chara*
Red algae – *Porphyra, Polysiphonia*

(97) **Answer :** (2)
Solution:
Leaves of conifers have thick cuticle.

(98) **Answer :** (2)
Hint:
Leaf blade is also known as lamina.
Solution:
In some leguminous plants, the leaf base may become swollen, which is called the pulvinus. The petiole help hold the blade to light. Long thin flexible petioles allow leaf blades to flutter in wind. The lamina or the leaf blade is the green expanded part of the leaf with veins and veinlets.

(99) **Answer :** (3)
Solution:
Racemose inflorescence shows acropetal succession, where younger flowers are present towards the apex and the older flowers are present at the base.

(100) **Answer :** (3)
Hint:
In basal type, placenta develops at the base of the ovary.
Solution:
In free central placentation, as in *Dianthus*, the ovules are borne on central axis and septa are absent in the ovary. In sunflower, basal type of placentation is observed.

(101) **Answer :** (1)
Hint:
In monocot stem, pericycle is absent.
Solution:
Pericycle consist of parenchymatous cells in roots.
The parenchymatous cells of pericycle become meristematic, and give rise to part of the vascular cambium and lateral roots in dicots.

(102) **Answer :** (4)
Solution:
Grass leaves have dumb-bell shaped guard cells.

(103) **Answer :** (3)
Solution:
The Golgi apparatus remains in close association with ER as the materials to be packaged in the forms vesicles from the ER fuses with the cis face of the Golgi apparatus.

(104) **Answer :** (2)
Solution:
At number of places the nuclear envelope is interrupted by minute pores, which are formed by the fusion of its two membranes.

(105) **Answer :** (2)
Solution:
Centrioles are absent in higher plants.

(106) **Answer :** (4)
Hint:
Cilia move particles in a specific direction.
Solution:
Ciliated epithelium lines the fallopian tubes (oviduct).

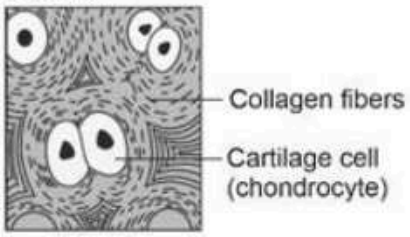
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(107) Answer : (3)

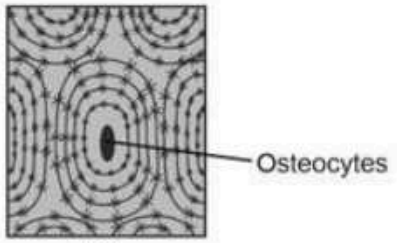
Hint:

Cartilage

Solution:



(a)



(b)

Lamellae are found in bones.

(108) Answer : (3)

Hint:

Lemon grass oils is an essential oil.

Solution:

Concanavalin A is a lectin and lemon grass oil is an essential oil.

(109) Answer : (3)

Solution:

Antibodies fight against infectious agents. In human body, five kinds of antibodies are found (IgA, IgG, IgM, IgD and IgE).

(110) Answer : (4)

Solution:

Collagen is the most abundant protein in the animal world and Ribulose Bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in the whole of the biosphere.

(111) Answer : (2)

Solution:

DNA, GTP and NAD possess pentose sugar. GTP stands for guanosine phosphate, and is composed of guanine, ribose and triphosphate groups.

(112) Answer : (4)

Solution:

Breathing is a mechanical process while respiration is a chemical/metabolic process.

(113) Answer : (3)

Solution:

Residual volume is the amount of air left in the lungs after maximal expiration.

(114) Answer : (1)

Hint:

Closure of tricuspid and bicuspid valves produces first heart sound.

Solution:

During each cardiac cycle, two prominent sounds are produced which can be easily heard through a stethoscope. The first heart sound (lub) is associated with the closure of the tricuspid and bicuspid valves whereas the second heart sound (dub) is associated with the closure of the semilunar valves. These sounds are of clinical diagnostic significance.

(115) Answer : (2)

Solution:

Heart failure means the state of heart when it is not pumping blood effectively enough to meet the needs of the body.

(116) Answer : (3)

Solution:

Nucleolus is found in eukaryotes.

(117) Answer : (1)

Solution:

Dissolution of synaptonemal complex – Diplotene
Formation of recombination nodule – Pachytene
Assembling of meiotic spindle at the poles – Diakinesis.

(118) Answer : (4)

Solution:

In telophase, reappearance of nucleolus, golgi complex and E.R. occur. Chromosomes lose their identity as discrete elements.

(119) Answer : (2)

Hint:

Bivalent is a complex formed by a pair of synapsed chromosome.

Solution:

Number of bivalent formed
= $\frac{1}{2}$
[total number of chromosomes]
= $\frac{1}{2}$
(42) = 21.

(120) Answer : (2)

Solution:

C_3 pathway only occurs in mesophyll cells.

(121) Answer : (3)

Solution:

The reaction centre chlorophyll *a* molecule in PS II has an absorption maxima at 680 nm.

(122) Answer : (1)

Solution:

Complete oxidation of acetyl CoA occurs in aerobic respiration through Krebs' cycle.

One turn of Krebs' cycle involves complete breakdown of acetyl CoA which produces 3NADH + H^+, 1 $FADH_2$ & 1 GTP.

$3NADH + H^+ = 3 \times 3 = 9ATP$, 1FADH2 = 2ATP, 1GTP will yield 1ATP.
Therefore, total 12ATP will be produced through complete oxidation of one molecule of acetyl coenzyme A.

(123) Answer : (4)

Solution:

CO_2 is not evolved during glycolysis.

(124) Answer : (2)

Hint:

Relative growth rate $= \frac{\text{Growth per unit time}}{\text{Initial size}} \times 100\%$

Solution:

Relative growth rate of leaves are as follows :

$A = \frac{10-5}{5} \times 100 = 100\%$

$B = \frac{20-15}{15} \times 100 = 33.33\%$

$C = \frac{15-10}{10} \times 100 = 50\%$

$D = \frac{22-20}{20} \times 100 = 10\%$

(125) Answer : (1)

Solution:

Synthetic auxins like 2, 4-D is used as weedicide and auxins also control xylem differentiation.

(126) Answer : (1)

Solution:

Pollen grain have prominent two layered wall. The hard outer layer is called exine made up of sporopollenin which is one of the most resistant organic materials known.
Pollen grain has prominent aperture called germ pores where sporopollenin is absent.

(127) Answer : (3)

Solution:

The proximal end of the filament is attached to thalamus or the petal of the flower. The process of formation of microspores from pollen mother cell through meiosis is called microsporogenesis.

(128) Answer : (4)
Solution:
Both autogamy and geitonogamy are prevented in dioecious plants.
Papaya is a dioecious plant.

(129) Answer : (2)
Solution:
True breeders are homozygous for a trait. Both the alleles in a zygote are contributed by both the parents.

(130) Answer : (3)
Solution:
In phenylketonuria, phenylalanine is not converted to tyrosine due to lack of enzyme phenylalanine hydroxylase.

(131) Answer : (3)
Solution:
Counter current mechanism helps to maintain a concentration gradient in the medullary interstitium. Presence of such interstitial gradient helps in an easy passage of water from the collecting tubules thereby concentrating the urine.

(132) Answer : (3)
Solution:
Bony fishes are usually ammonotelic.

(133) Answer : (3)
Hint:
Blood capillaries of medulla are vasa recta.
Solution:
The loop of Henle of cortical nephrons do not reach deep into pyramids. Vasa recta are well developed in JG nephrons.

(134) Answer : (1)
Solution:
A – Cartilage present in the body
B – Is a bony feature

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(135) Answer : (4)
Solution:
The sequence of events for muscle contraction is
(i) Nerve impulse reaches neuro-muscular junction
(ii) Impulse stimulates sarcoplasmic reticulum
(iii) Release of calcium ions
(iv) Cross-bridge formation
(v) Muscle contraction

(136) Answer : (4)
Solution:
Muscular dystrophy is a genetic disorder due to defects in the gene encoding dystrophin protein while rheumatoid arthritis is an auto-immune disorder that affects the synovial lining of joints.

(137) Answer : (1)
Hint:
Synapses are junctions through which a nerve impulse is transmitted from one neuron to another.
Solution:
Chemical synapses possess larger synaptic clefts compared to electrical synapses.
Hence, impulse transmission is always faster in electrical synapses than chemical synapses.

(138) Answer : (3)
Solution:
Angiotensin II is considered as a proteinaceous hormone which increases blood pressure due to its potent vasoconstrictory effect on blood vessels. ANF decreases blood pressure by increasing GFR and causing vasodilation. Aldosterone increases blood volume, blood pressure and GFR but it is steroidal in nature. Thyroxine is not a proteinaceous hormone.

(139) Answer : (1)
Solution:
Glucocorticoids such as cortisol exhibits immunosuppression and anti-inflammatory effect. It stimulates gluconeogenesis, proteolysis and lipolysis. It inhibits the utilisation of amino acids.

(140) Answer : (2)
Solution:

Hydra is a fresh water polyp and has tentacles for locomotion. Cnidoblasts are present on tentacles and on the body. It has a central gastrovascular cavity with a single opening, mouth on hypostome.

(141) Answer : (4)
Solution:
No mouth and alimentary canal are present in tape worms.
Tape worms are pseudometamerically segmented.
Hooks and suckers help in attachment.

(142) Answer : (3)
Solution:
Chelone, *Bufo* and *Hyla* are oviparous.
Great white shark is viviparous and aquatic.
Exocoetus, *Betta*, *Clarias*, *Catla* are oviparous and aquatic gnathostomes(bony fish).
Balaenoptera is an aquatic viviparous mammal.

(143) Answer : (1)
Solution:
Frogs are beneficial for mankind because they eat insects and protect the crop. Frogs maintain ecological balance, because these serve as an important link of food chain and food web in the ecosystem.
In some countries, the muscular legs of frogs are used as food by man.

(144) Answer : (2)
Solution:
Hypopharynx, labium, labrum, frons are unpaired structures.

(145) Answer : (2)
Solution:
The amnion present around mammalian embryo is derived from ectoderm and mesoderm.

(146) Answer : (3)
Solution:
In humans, insemination occurs before the fusion of gametes in the fallopian tube.

(147) Answer : (2)
Solution:

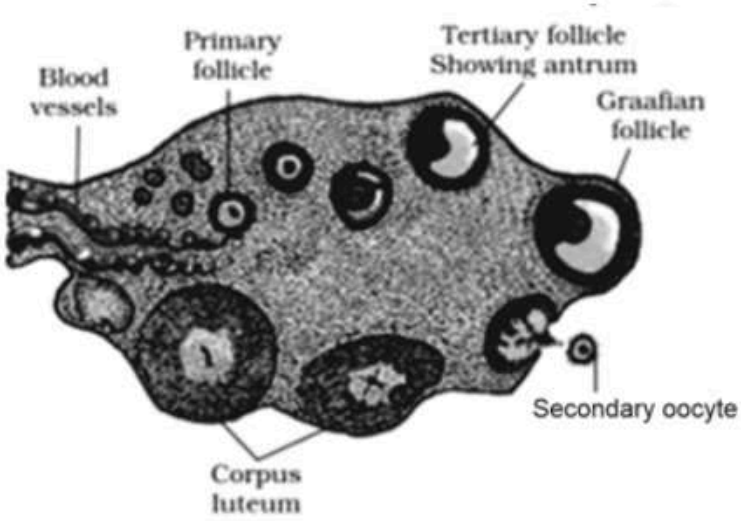
Insemination	–	Transfer of sperms into the female genital tract
Implantation	–	Attachment of blastocyst to uterine wall
Gestation	–	Embryonic development
Parturition	–	Delivery of the baby

(148) Answer : (3)
Hint:
12 menstrual cycles.
Solution:
Reproductive life of female = 16 years
Primary polar bodies = 16 × 12 months = 192
Primary polar bodies
= 192 – [9 months of pregnancy + 6 months of lactation]
= 192 – 15= 177
Secondary polar body = 1 during fertilization
Total polar bodies = 177 + 1= 178

(149) Answer : (1)
Solution:
The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus, *etc.*, form a complex structure called the limbic lobe or limbic system. Along with the hypothalamus, it is involved in the regulation of sexual behaviour, expression of emotional reactions (*e.g.*, excitement, pleasure, rage and fear), and motivation.

(150) Answer : (2)
Solution:
Graafian follicle ruptures to release ovum.

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(151) Answer : (1)

Hint:

Reversibility is very poor.

Solution:

Surgical methods of contraception are also called sterilisation. Sterilisation in males is called vasectomy in which a part of vas deferens is removed or tied up. Sterilisation in females is called tubectomy in which a small part of fallopian tube is removed or tied up. Sterilisation has high contraceptive efficiency but reversibility is very poor.

(152) Answer : (2)

Solution:

While copper ions primarily reduce sperm motility, their secondary effect is altering biochemical environment of the endometrium. This prevents a fertilized egg (blastocyst) from successfully implanting.

(153) Answer : (3)

Solution:

Stabilising selection – the peak of bell-shaped graph gets higher and narrower.

Directional/Progressive selection – peak shifts in the direction which is selected by the nature.

(154) Answer : (2)

Solution:

$q^2 = \frac{98}{200} = 0.49$

$q = \sqrt{0.49} = 0.7$

$p + q = 1$

$p = 0.3$

$p^2 = 0.09 = 9\%$

(155) Answer : (1)

Hint:

Antibody present in body fluids.

Solution:

Colostrum is a yellowish liquid substance that a nursing mother produces from her breasts 25-48 hours after delivery which is very important in the transfer of antibodies to the newborn.

(156) Answer : (4)

Hint:

In this mutation, there is change in linear order of genes.

Solution:

Change in linear order of genes by rotation of a section of chromosome by 180° is called inversion mutation. The inversion shown in the figure involves centromere.

(157) Answer : (1)

Solution:

ZW – ZZ type of sex determination is found in birds.

Hens are heterogametic and thus, they produce two types of eggs, *i.e.*, (A + Z) and (A + W). Therefore, females are responsible to determine the sex of the chicks.

(158) Answer : (4)

Solution:

In first generation only hybrid DNA is obtained. When the hybrid DNA is replicated in ^{14}N medium, one light and one hybrid DNA will be synthesised from each hybrid DNA. There will be no synthesis of heavy DNA. In this way, only 50 % of DNA would be hybrid and 50 % would be light.

(159) Answer : (3)
Solution:
Genetic code is degenerate because some amino acids are coded by more than one codons. Given statement does not provide any idea about degeneracy of codons.

(160) Answer : (3)
Solution:
Satellite DNA form a large portion of human genome. Less than 2% of human genome codes for proteins. Satellite DNA normally does not code for any protein.

(161) Answer : (4)
Solution:
AUG, UGG, UGC, UAA, U

(162) Answer : (2)
Solution:
Avery, MacLeod and McCarty discovered that proteases and RNases did not affect transformation.

(163) Answer : (3)
Solution:
Another bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*.

(164) Answer : (4)
Solution:
The main sources of biofertilisers are bacteria, fungi and cyanobacteria.

(165) Answer : (4)
Hint:
The given age pyramid is urn shaped.
Solution:
Fewer individuals in pre-reproductive group indicates that the population size is declining with time.

(166) Answer : (3)
Solution:
1. Not all orchids offer rewards.
2. *Ophrys* takes the help of 'pseudo-copulation' to get pollination done by a species of bee.

(167) Answer : (3)
Solution:
Gross primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis.
Net primary productivity is the available biomass for the consumption to herbivores and decomposers.

(168) Answer : (4)
Solution:
Stratification is an structural aspect of ecosystem

(169) Answer : (1)
Solution:
Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable. Such constant environments promote niche specialisation and lead to a greater species diversity.

(170) Answer : (4)
Solution:
National park is an *in-situ* conservation strategy.

(171) Answer : (3)
Solution:
Malignant tumors undergo metastasis, whereas benign tumors confined to their original location.

(172) Answer : (4)
Hint:
Pneumonia causing pathogens can be bacteria, viruses or fungi.
Solution:
In pneumonia, the alveoli get filled with fluid leading to severe problems in respiration. Symptoms of pneumonia include fever, chills, cough and headache.

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(173) Answer : (1)

Solution:

If a circular DNA is cut at 6 restriction sites, it will produce 6 fragments.

(174) Answer : (4)

Hint:

Haemophilus influenzae is the microbial origin of this enzyme.

Solution:

Hind II was the first discovered type II restriction endonuclease.

(175) Answer : (1)

Hint:

Series of processes before marketing as a finished product.

Solution:

After completion of biosynthetic stage, downstream processing includes separation and purification.

(176) Answer : (4)

Solution:

*Eco*RI cuts the DNA between bases G and A only when the sequence GAATTC is present in the DNA.

(177) Answer : (1)

Hint:

Bacterial cells must be made 'competent' to take up DNA.

Solution:

The methods for insertion of recombinant DNA into the host cell/organism include microinjection, biolistics, *etc*.

(178) Answer : (4)

Solution:

RNAi takes place in all eukaryotic organisms as a method of cellular defence. dsRNA initiates RNAi that leads to gene silencing.

(179) Answer : (3)

Solution:

GEAC (Genetic Engineering Approval Committee) make decisions regarding validity of GM research and the safety of introducing GM-organisms for public services. RNAi involves silencing of specific mRNA by forming dsRNA, so it is a post transcriptional method.

(180) Answer : (4)

Solution:

· ELISA can be used for the early detection of an infection either by detecting the presence of pathogenic antigens or by detecting the antibodies synthesized against the pathogen.

· Southern blotting is used to detect a specific DNA sequence in the given sample. One can use this technique to detect the presence of pathogenic DNA.

· In hybridization technique, a ssDNA/ssRNA tagged with a radioactive molecule (probe) is allowed to hybridize to its complementary DNA in a clone of cells followed by detection using autoradiography. It is used to find a mutated gene.

· Western blotting technique is used to detect a specific protein molecule in a mixture of proteins.